

SUPPLY CHAIN AND LOGISTICS MANAGEMENT

VI SEMESTER

CORE COURSE

BBA6 B16

(Marketing Elective IV)

B.B.A

(2019 Admission onwards)



UNIVERSITY OF CALICUT

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UNIVERSITY OF CALICUT

SCHOOL OF DISTANCE EDUCATION

Study Material

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**SUPPLY CHAIN AND LOGISTICS
MANAGEMENT**

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Syllabus
BBA6B16 (Marketing Elective 4)

SUPPLY CHAIN AND LOGISTICS MANAGEMENT

Time: 5 Hours per week

Credits: 4

Internal: 20

External: 80

Objectives:

- To impart knowledge and understanding to students on Supply Chain Management and its relevance to today's business decisionmaking.
- To gain the knowledge of possibilities of efficient optimization and management of operation in Logistics Management and also the ability to apply them in the enterprisereality.
- **Learning Outcome:**
- The students will be able to examine the details of planning and control processes in logistics management. The student at the end of the course will be able to understand the various logistics operations and their importance in improving the business

Module I : Supply Chain: definition – Objectives – Key benefits of SCM– SCM process- SCM process flows– Material flow, information and money flow– Major components of flow of supply chain- Transport, warehouse, sourcing and procurement, returns, post sales service – SCM decisions and skills – Strategy formulation in SCM – Value in Supply Chain.

10 Hours

Module II: Strategic Sourcing: Meaning – Steps in strategic sourcing – Supply chain collaboration Meaning, Types and benefit of Supply chain collaboration – Role of inventory in SCM- Inventory management techniques in supply chain: Stock review, just in time and ABC analysis.

15Hours

Module III : Transportation Selection – Tradeoff – modes of transportation – models for transportation and distribution – factors affecting network effectiveness – 3 PL advantages – Indian transport infrastructure – IT solutions – EDI, e-Commerce, e-Procurement – Bar Coding and RFID technology

15 Hours

UNIT IV : Logistics Management: The Logistics of Business – The Logistical Value Proposition – The Work of Logistics – Logistical Operating Arrangements – Flexible Structure – Supply Chain Synchronization.. Transport Functionality, Principles and Participants – Transportation Service – Transportation Economics and Pricing – Transport Administration – Documentation

20Hours

UNIT V : International Logistics and Supply Chain Management: Meaning and objectives – importance in global economy– Characteristics of global supply chains– Global Supply Chain Integration – Supply Chain Security – International Sourcing – Role of Government in controlling international trade and its impact on Logistics and Supply Chain.

20 Hours

Reference books:

1. Sahay B.S, *Supply Chain Management for Global Competitiveness*, Macmillan India Ltd., NewDelhi.
2. Reguram G, Rangaraj N, *Logistics and Supply Chain Management Cases and Concepts*, Macmillan India Ltd., NewDelhi.
3. Bowersox, *Logistical Management*, Mc-GrawHill
4. D K Agrawal, 'Distribution and Logistics Management: A Strategic MarketingApproach', Macmillan publishers India2007

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4	Logistics Management	100 – 135
5	International Logistics and Supply Chain Management	136 - 153

Module I

SUPPLY CHAIN

The practice of supply chain management is guided by some basic underlying concepts that have not changed much over the centuries. Several hundred years ago, Napoleon made the remark, “An army marches on its stomach.” Napoleon was a master strategist and a skillful general and this remark shows that he clearly understood the importance of what we would now call an efficient supply chain. Unless the soldiers are fed, the army cannot move.

Along these same lines, there is another saying that goes, “Amateurs talk strategy and professionals talk logistics.” People can discuss all sorts of grand strategies and dashing maneuvers but none of that will be possible without first figuring out how to meet the day-to-day demands of providing an army with fuel, spare parts, food, shelter, and ammunition. It is the seemingly mundane activities of the quartermaster and the supply sergeants that often determine an army’s success. This has many analogies in business.

The term “supply chain management” arose in the late 1980s and came into widespread use in the 1990s. Prior to that time, businesses used terms such as “logistics” and “operations management” instead. Some definitions of a supply chain are offered below:

- “A supply chain is the alignment of firms that bring products or services to market.”
- “A supply chain consists of all stages involved, directly or indirectly, in fulfilling a customer request. The supply chain not only includes the manufacturer and suppliers, but also transporters, warehouses, retailers, and customers themselves.”— Chopra and Meindl

- “A supply chain is a network of facilities and distribution options that performs the functions of procurement of materials, transformation of these materials into intermediate and finished products, and the distribution of these finished products to customers.”— Ganeshan and Harrison

The definition of supply chain management developed and used by The Global

Supply Chain Forum: “Supply Chain Management is the integration of key business processes from end user through original suppliers that provides products, services, and information that add value for customers and other stakeholders.”

Supply Chains

Networks of manufacturers and service providers that work together to move goods from the raw material stage through to the end user linked through physical, information, and monetary flows

Within the organization, the supply chain refers to a wide range of functional areas. These include Supply Chain Management-related activities such as inbound and outbound transportation, warehousing, and inventory control. Sourcing, procurement, and supply management fall under the supply-chain umbrella, too. Forecasting, production planning and scheduling, order processing, and customer service all are part of the process as well.

Importantly, it also embodies the information systems so necessary to monitor all of these activities.

Simply stated, *"the supply chain encompasses all of those activities associated with moving goods from the raw-materials stage through to the end user."*

Advocates for this business process realized that significant productivity increases could only come from

managing relationships, information, and material flow across enterprise borders. One of the best definitions of supply-chain management offered to date comes from Bernard J. (Bud) LaLonde, professor emeritus of Supply Chain Management at Ohio State University. LaLonde defines supply-chain management as follows: "The delivery of enhanced customer and economic value through synchronized management of the flow of physical goods and associated information from sourcing to consumption." As the "from sourcing to consumption" part of definition suggests, though, achieving the real potential of supply-chain management requires integration--not only of these entities within the organization, but also of the external partners. The latter include the suppliers, distributors, carriers, customers, and even the ultimate consumers. All are central players in what James E. Morehouse of A. T. Kearney calls the extended supply chain. "The goal of the extended enterprise is to do a better job of serving the ultimate consumer," Superior service, he continues, leads to increased market share. Increased share, in turn, brings with it competitive advantages such as lower warehousing and transportation costs, reduced inventory levels, less waste, and lower transaction costs.

The customer is the key to both quantifying and communicating the supply chain's value, confirms Shrawan Singh, vice president of integrated supply-chain management at Xerox.

"If you can start measuring customer satisfaction associated with what a supply chain can do for a customer and also link customer satisfaction in terms of profit or revenue growth," Singh explains, "then you can attach customer values to profit & loss and to the balance sheet." The Supply Chain Management Program integrates topics from manufacturing operations, purchasing, transportation,

and physical distribution into a unified program. Successful supply chain management, then, coordinates and integrates all of these activities into a seamless process. It embraces and links all of the partners in the chain. In addition to the departments within the organization, these partners include vendors, carriers, third party companies, and information systems providers.

Supply Chain Management is primarily concerned with the efficient integration of suppliers, factories, warehouses and stores so that merchandise is produced and distributed in the right quantities, to the right locations and at the right time, and so as to minimize total system cost subject to satisfying service requirements

Importance of Supply Chain Management

The importance of supply chain management decisions are as follows

- **Helps in achieving success:** companies being a leader at using supply chain design, planning and operation help in achieving success.
- **Effective flow of goods and information:** MNCs and other giant companies have invested heavily in transportation and information infrastructure helps in achieving effective flow of goods and information.
- **Reduces the level of inventory with the manufacturer:** centralized manufacturing and inventory in a few locations makes the companies to produce and supply large variety of products at low inventory levels.

Supply Chain Management Today

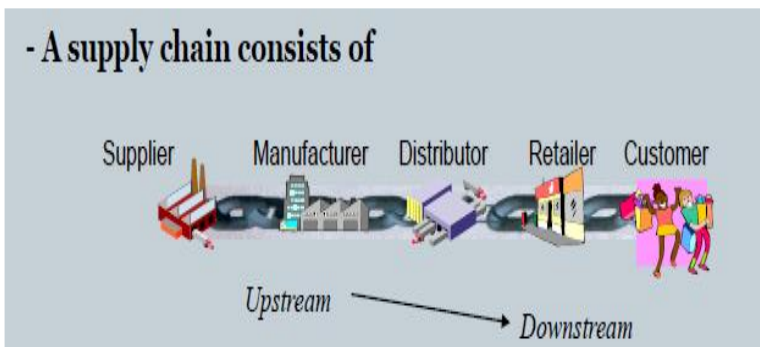
If we take the view that Supply Chain Management is what Supply Chain Management people do, then in 1997

Supply Chain Management has a firm hand on all aspects of physical distribution and materials management. Seventy-five percent or more of respondents included the following activities as part of their company's Supply Chain Management department functions:

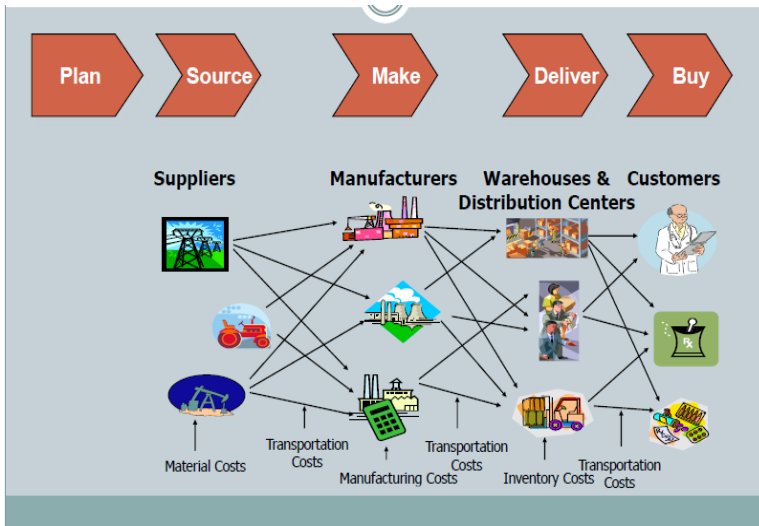
- ✓ Inventory management
- ✓ Transportation service procurement
- ✓ Materials handling
- ✓ Inbound transportation
- ✓ Transportation operations management
- ✓ Warehousing management
- ✓

Moreover, the Supply Chain Management department is expected to increase its range of responsibilities, most often in line with the thinking that sees the order fulfillment process as one co-ordinate set of activities. Thus the functions most often cited as planning to formally include in the Supply Chain Management department are:

- ✓ Customer service performance monitoring
- ✓ Order processing/customer service
- ✓ Supply Chain Management budget forecasting



The supply chain – an overview



In nutshell Supply Chain Management (SCM) is -

A set of approaches used to efficiently integrate

- Suppliers
- Manufacturers
- Warehouses
- Distribution centers

So that the product is produced and distributed

- In the right quantities
- To the right locations
- And at the right time

- System-wide costs are minimized and
- Service level requirements are satisfied

The Objective of a Supply Chain Management

Supply Chain Management becomes a tool to help accomplish corporate strategic objectives:

- Reducing working capital,
- Taking assets off the balance sheet,
- Accelerating cash-to-cash cycles,
- Reduce costs
- Minimize delays
- Maximize overall value created
- Increasing inventory turns.

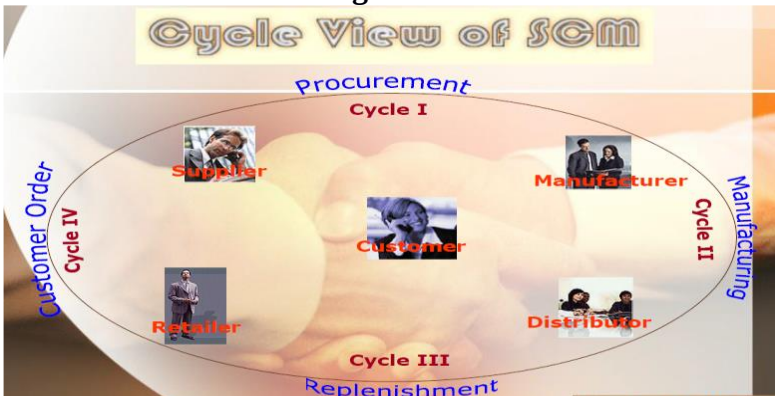
Process View of a Supply Chain

❖ Cycle view of a Supply Chain

Processes in a supply chain are divided into a series of cycles, each performed at the interfaces between two successive supply chain stages. Each cycle occurs at the interface between two successive stages. Cycle view clearly defines processes involved and the owners of each process. Specifies the roles and responsibilities of each member and the desired outcome of each process (see below figure 1.3)

- Customer order cycle (customer-retailer)
- Replenishment cycle (retailer-distributor)
- Manufacturing cycle (distributor-manufacturer)
- Procurement cycle (manufacturer-supplier).

Figure1.3



❖ Push/pull view of a Supply Chain

Processes in a supply chain are divided into two categories depending on whether they are executed in response to a customer order (pull) or in anticipation of a customer order (push) (see below figure 1.4)

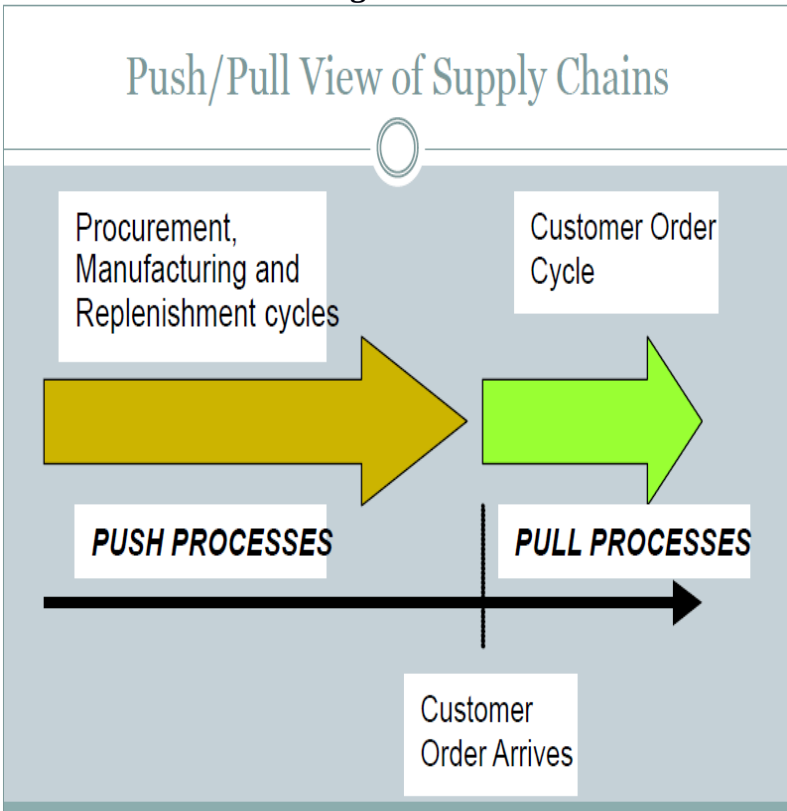
Pull:(execution is initiated in response to a customer order (reactive). In a pull-based supply chain, manufacturing is demand driven so that it is coordinated with actual external customer demand rather than a forecast Lead-time reduction occurs as the variabilities are better monitored in pull-based SCM. Pull-based systems are often difficult to implement when lead times are so long that it is impractical to react to demand information

Push: (execution is initiated in anticipation of customer orders (speculative)). A push-based SCM takes longer to react to the changing market place In a push-based supply chain, production decisions are usually based on long-term forecasts In these

strategies, SCM experience increased transportation costs, high inventory levels and high manufacturing costs

Push/Pull View of Supply Chain Processes is useful in considering strategic decisions relating to supply chain design – more global view of how supply chain processes relate to customer orders. The relative proportion of push and pull processes can have an impact on supply chain performance

Figure1.4



Characteristics of Push and Pull strategies

	PUSH	PULL
Objective	Minimize Cost	Maximize Service Level
Complexity	High	Low
Focus	Resource Allocation	Responsiveness
Lead Time	Long	Short
Processes	Supply Chain Planning	Order Fulfillment

Supply-Chain Principles

If supply-chain management has become top management's new "religion," then it needs a doctrine. Andersen Consulting has stepped forward to provide the needed guidance, espousing what it calls the "Seven Principles" of supply-chain management. When consistently and comprehensively followed, the consulting firm says, these seven principles bring a host of competitive advantages. *The seven principles as articulated by Andersen Consulting are as follows:*

- 1. Segment customers based on service needs.** Companies traditionally have grouped customers by industry, product, or trade channel and then provided the same level of service to everyone within a segment. Effective supply-chain management, by contrast, groups customers by distinct service needs--regardless of industry--and then tailors services to those particular segments.

2. Customise the Supply Chain Management network.

In designing their Supply Chain Management network, companies need to focus intensely on the service requirements and profitability of the customer segments identified. The conventional approach of creating a "monolithic" Supply Chain Management network runs counter to successful supply-chain management.

3. Listen to signals of market demand and plan accordingly.

Sales and operations planning must span the entire chain to detect early warning signals of changing demand in ordering patterns, customer promotions, and so forth. This demand-intensive approach leads to more consistent forecasts and optimal resource allocation.

4. Differentiate product closer to the customer.

Companies today no longer can afford to stockpile inventory to compensate for possible forecasting errors. Instead, they need to postpone product differentiation in the manufacturing process closer to actual consumer demand.

5. Strategically manage the sources of supply.

By working closely with their key suppliers to reduce the overall costs of owning materials and services, supply-chain management leaders enhance margins both for themselves and their suppliers. Beating multiple suppliers over the head for the lowest price is out, Andersen advises. "Gain sharing" is in.

6. Develop a supply-chain-wide technology strategy.

As one of the cornerstones of successful supply-chain management, information technology must support multiple levels of decision making. It also should afford a clear view of the flow of products, services, and information.

7. Adopt channel-spanning performance measures.

Excellent supply-chain measurement systems do more than

just monitor internal functions. They adopt measures that apply to every link in the supply chain. Importantly, these measurement systems embrace both service and financial metrics, such as each account's true profitability.

The principles are not easy to implement, the Andersen consultants say, because they run counter to ingrained functionally oriented thinking about how companies organise, operate, and serve customers. The organisations that do persevere and build a successful supply chain have proved convincingly that you can please customers and enjoy growth by doing so.

Participants in the Supply Chain

In its simplest form, a supply chain is composed of a company and the suppliers and customers of that company. This is the basic group of participants that creates a simple supply chain. Extended supply chains contain three additional types of participants. First there is the supplier's supplier or the ultimate supplier at the beginning of an extended supply chain. Then there is the customer's customer or ultimate customer at the end of an extended supply chain. Finally there is a whole category of companies who are service providers to other companies in the supply chain. These are companies who supply services in logistics, finance, marketing, and information technology.

In any given supply chain there is some combination of companies who perform different functions. There are companies that are producers, distributors or wholesalers, retailers, and companies or individuals who are the customers, the final consumers of a product. Supporting these companies there will be other companies that are service providers that provide a range of needed services.

❖ **Producers**

Producers or manufacturers are organizations that make a product. This includes companies that are producers of raw materials and companies that are producers of finished goods. Producers of raw materials are organizations that mine for minerals, drill for oil and gas, and cut timber. It also includes organizations that farm the land, raise animals, or catch seafood. Producers of finished goods use the raw materials and sub assemblies made by other producers to create their products.

Producers can create products that are intangible items such as music, entertainment, software, or designs. A product can also be a service such as mowing a lawn, cleaning an office, performing surgery, or teaching a skill. In many instances the producers of tangible, industrial products are moving to areas of the world where labor is less costly. Producers in the developed world of North America, Europe, and parts of Asia are increasingly producers of intangible items and services.

❖ **Distributors**

Distributors are companies that take inventory in bulk from producers and deliver a bundle of related product lines to customers. Distributors are also known as wholesalers. They typically sell to other businesses and they sell products in larger quantities than an individual consumer would usually buy. Distributors buffer the producers from fluctuations in product demand by stocking inventory and doing much of the sales work to find and service customers. For the customer, distributors fulfill the “Time and Place” function—they deliver products when and where the customer wants them. A distributor is typically an organization that takes ownership of significant inventories of products that they buy from producers and sell to

consumers. In addition to product promotion and sales, other functions the distributor performs are inventory management, warehouse operations, and product transportation as well as customer support and post-sales service. A distributor can also be an organization that only brokers a product between the producer and the customer and never takes ownership of that product. This kind of distributor performs mainly the functions of product promotion and sales. In both these cases, as the needs of customers evolve and the range of available products changes, the distributor is the agent that continually tracks customer needs and matches them with products available.

❖ **Retailers**

Retailers stock inventory and sell in smaller quantities to the general public. This organization also closely tracks the preferences and demands of the customers that it sells to. It advertises to its customers and often uses some combination of price, product selection, service, and convenience as the primary draw to attract customers for the products it sells. Discount department stores attract customers using price and wide product selection. Upscale specialty stores offer a unique line of products and high levels of service. Fast food restaurants use convenience and low prices as their draw.

❖ **Customers**

Customers or consumers are any organization that purchases and uses a product. A customer organization may purchase a product in order to incorporate it into another product that they in turn sell to other customers or a customer may be the final end user of a product who buys the product in order to consume it.

❖ **Service Providers**

These are organizations that provide services to producers, distributors, retailers, and customers. Service providers have developed special expertise and skills that focus on a particular activity needed by a supply chain. Because of this, they are able to perform these services more effectively and at a better price than producers, distributors, retailers, or consumers could do on their own. Some common service providers in any supply chain are providers of transportation services and warehousing services. These are trucking companies and public warehouse companies and they are known as logistics providers. Financial service providers deliver services such as making loans, doing credit analysis, and collecting on past due invoices. These are banks, credit rating companies, and collection agencies.

Some service providers deliver market research and advertising, while others provide product design, engineering services, legal services, and management advice. Still other service providers offer information technology and data collection services. All these service providers are integrated to a greater or lesser degree into the ongoing operations of the producers, distributors, retailers, and consumers in the supply chain.

Supply chains are composed of repeating sets of participants that fall into one or more of these categories. Over time the needs of the supply chain as a whole remain fairly stable. What changes is the mix of participants in the supply chain and the roles that each participant plays.

In some supply chains, there are few service providers because the other participants perform these services on their own. In other supply chains very efficient providers of specialized services have evolved and the other

participants outsource work to these service providers instead of doing it themselves. Examples of supply chain structure are shown as –

❖ **Franchisers.**

Franchisers are owners of business systems and processes who grant one or more third parties (franchisees) the right to use their business systems or processes, as well as **trademarks** or trade names to produce and market goods (or services) according to uniform specifications in exchange for a one-time franchise fee plus a percentage of sales revenue (royalty).

❖ **Sales representatives.**

Sales representatives market, advertises, promote and solicit the sale of the goods on behalf of the seller (such as a manufacturer or distributor) to the seller's customers in the specified territory. Sales representatives do not take title to the goods or bear inventory risk or risk of loss regarding the goods. They also do not bear the credit risk of the customers.

❖ **Logistics providers.**

These entities provide a variety of services on behalf of other participants in the supply chain to move the goods between the participants. Logistics providers may take temporary custody of the goods, but do not take title to the goods. Logistics providers include:

- a) Ware housemen, which are entities engaged in the business of storing goods for hire;
- b) Carriers, which are entities like trucking companies that issue bills of lading; and

c) Customs brokers, which are entities engaged in the business of clearing goods through customs barriers for importers and exporters.

❖ **Financiers.**

In addition to sellers who provide seller-financing, such as extended or deferred payment terms, these entities include banks, factoring companies and other entities that provide:

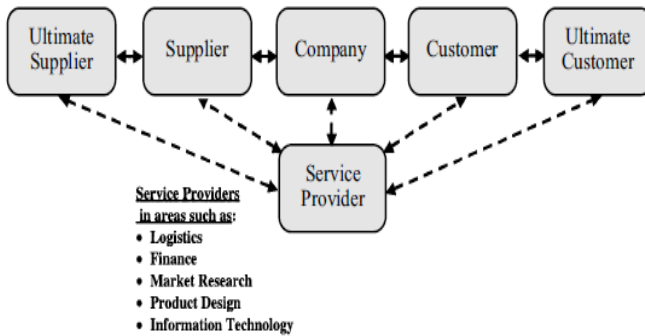
- a) Purchase-money financing for a buyer to pay the purchase price of goods;
- b) Commercial letters of credit to buyers to further the payment of goods in the ordinary course of a transaction; or
- c) Factoring to sellers who sell or assign their ***receivables*** to accelerate their cash flow.

Supply Chain Structure

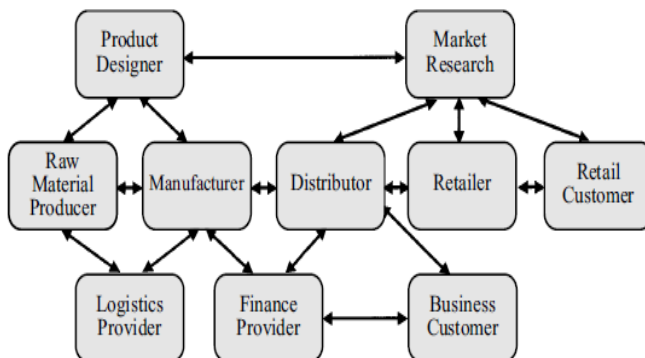
Simple Supply Chain



Extended Supply Chain



Example of an Extended Supply Chain



SUPPLY CHAIN PLANNING

Supply Chain Planning enables manufacturers to synchronize enterprise-wide production and supply with enterprise-wide demand. The solution allows manufacturers to aggregate total demand and centrally plan for the production capacity and supplies required to satisfy that demand. Supply Chain Planning consolidates sales, production, inventory and purchasing information to help companies become more demand-driven and actually manufacture items based on real demand.

In today's demand-driven market, it is critical for manufacturers to optimize and integrate sales and logistics and incorporate such data into the production schedule in a timely manner.

Supply Chain Planning delivers substantial benefits to manufacturers including:

1. Increased responsiveness to market changes
2. Improved visibility into aggregated demand as well as enterprise-wide production and supply
3. Reduced inventory levels
4. Improved customer service and on-time delivery performance
5. Optimized supply to meet demand profitably
6. Lowered inventory, distribution and transportation costs
7. Increased demand forecast accuracy with compressed planning cycle times

SUPPLY CHAIN MANAGEMENT PROCESS

The Global Supply Chain Forum identified eight key processes that make up the core of supply chain management:

- Customer Relationship Management
- Customer Service Management
- Demand Management
- Order Fulfillment
- Manufacturing Flow Management
- Procurement (supplier relationship management)
- Product Development and Commercialization
- Returns (returns management).

The eight key business processes run the length of the supply chain and cut across firms and functional silos within each firm. Functional silos include Marketing, Research and Development, Finance, Production, Purchasing and Logistics. Activities in these processes reside inside a functional silo, but an entire process will not be contained within one function. Each process is described at strategic and operational levels.

The strategic portion consists of the establishment and strategic management of each process, and provides a blueprint for implementation.

This is a necessary first step in integrating the firm with other members of the Supply chain. The operational portion is the actualization of the process once it has been established.

DECISION PHASES IN A SUPPLY CHAIN

Successful supply chain management requires many decisions relating to the flow of information, product, and funds. Each decision should be made to raise the supply chain surplus. These decisions fall into three categories or phases, depending on the frequency of each decision and the time frame during which a decision phase has an impact. As a result, each category of decisions must consider uncertainty over the decision horizon.

1. ***Supply Chain Strategy or Design:*** During this phase, given the marketing and pricing plans for a product, a company decides how to structure the supply chain over the next several years. It decides what the chain's configuration will be, how resources will be allocated, and what processes each stage will perform. Strategic decisions made by companies include whether to outsource or perform a supply chain function in-house, the location and capacities of production and warehousing facilities, the products to be manufactured or stored at various locations, the modes of transportation to be made available along different shipping legs, and the type of information system to be utilized.

A firm must ensure that the supply chain configuration supports its strategic objectives and increases the supply chain surplus during this phase. Cisco's decisions regarding its choice of supply sources for components, contract manufacturers for manufacturing, and the location and capacity of its warehouses, are all supply chain design or strategic decisions. Supply chain design decisions are typically made for the long term (a matter of years) and are very expensive to alter on short notice. Consequently, when companies make these decisions, they must take into account uncertainty in anticipated market conditions over the next few years.

2. **Supply Chain Planning:** For decisions made during this phase, the time frame considered is a quarter to a year. Therefore, the supply chain's configuration determined in the strategic phase is fixed. This configuration establishes constraints within which planning must be done. The goal of planning is to maximize the supply chain surplus that can be generated over the planning horizon given the constraints established during the strategic or design phase. Companies start the planning phase with a forecast for the coming year (or a comparable time frame) of demand in different markets.

Planning includes making decisions regarding which markets will be supplied from which locations, the subcontracting of manufacturing, the inventory policies to be followed, and the timing and size of marketing and price promotions. Dell's decisions regarding markets supplied by a production facility and target production quantities at each location are classified as planning decisions. Planning establishes parameters within which a supply chain will function over a specified period of time. In the planning phase, companies must include uncertainty in demand, exchange rates, and competition over this time horizon in their decisions. Given a shorter time frame and better forecasts than the design phase, companies in the planning phase try to incorporate any flexibility built into the supply chain in the design phase and exploit it to optimize performance. As a result of the planning phase, companies define a set of operating policies that govern short-term operations.

3. **Supply Chain Operation:** The time horizon here is weekly or daily, and during this phase companies make decisions regarding individual customer orders. At the operational level, supply chain configuration is considered fixed, and planning policies are already defined. The goal

of supply chain operations is to handle incoming customer orders in the best possible manner. During this phase, firms allocate inventory or production to individual orders, set a date that an order is to be filled, generate pick lists at a warehouse, allocate an order to a particular shipping mode and shipment, set delivery schedules of trucks, and place replenishment orders. Because operational decisions are being made in the short term (minutes, hours, or days), there is less uncertainty about demand information. Given the constraints established by the configuration and planning policies, the goal during the operation phase is to exploit the reduction of uncertainty and optimize performance.

The design, planning, and operation of a supply chain have a strong impact on overall profitability and success. It is fair to state that a large part of the success of firms like Wal-Mart and Dell can be attributed to their effective supply chain design, planning, and operation.

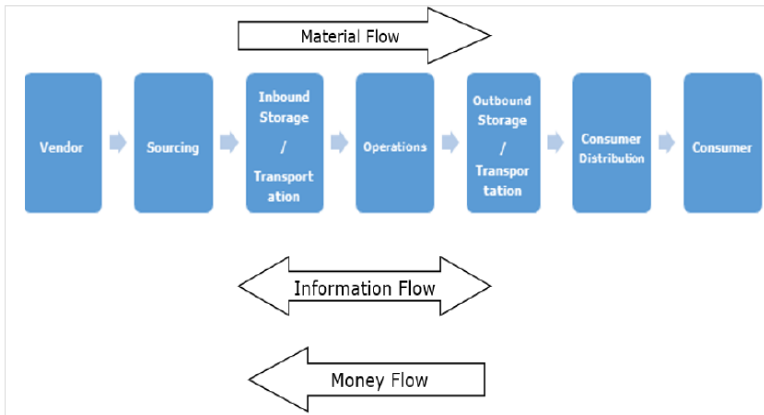
SCM – PROCESS FLOW

Supply chain management can be defined as a systematic flow of materials, goods, and related information among suppliers, companies, retailers, and consumers.

Types

There are three different types of flow in supply chain management:

- Material flow
- Information/Data flow
- Money flow



Let us consider each of these flows in detail and also see how effectively they are applicable to Indian companies.

Material Flow

Material flow includes a smooth flow of an item from the producer to the consumer. This is possible through various warehouses among distributors, dealers and retailers.

The main challenge we face is in ensuring that the material flows as inventory quickly without any stoppage through different points in the chain. The quicker it moves, the better it is for the enterprise, as it minimizes the cash cycle.

The item can also flow from the consumer to the producer for any kind of repairs, or exchange for an end of life material. Finally, completed goods flow from customers to their consumers through different agencies. A process known as 3PL is in place in this scenario. There is also an internal flow within the customer company.

Information Flow

Information/data flow comprises the request for quotation, purchase order, monthly schedules, engineering change requests, quality complaints and reports on supplier performance from customer side to the supplier.

From the producer's side to the consumer's side, the information flow consists of the presentation of the company, offer, confirmation of purchase order, reports on action taken on deviation, dispatch details, report on inventory, invoices, etc.

For a successful supply chain, regular interaction is necessary between the producer and the consumer. In many instances, we can see that other partners like distributors, dealers, retailers, logistic service providers participate in the information network.

In addition to this, several departments at the producer and consumer side are also a part of the information loop. Here we need to note that the internal information flow with the customer for in-house manufacture is different.

Money Flow

On the basis of the invoice raised by the producer, the clients examine the order for correctness. If the claims are correct, money flows from the clients to the respective producer. Flow of money is also observed from the producer side to the clients in the form of debit notes.

In short, to achieve an efficient and effective supply chain, it is essential to manage all three flows properly with minimal efforts. It is a difficult task for a supply chain manager to identify which information is critical for

decision-making. Therefore, he or she would prefer to have the visibility of all flows on the click of a button.

SCM – Flow Components

After understanding the basic flows involved in the supply chain management, we need to consider the different elements present in this flow. Thus, the different components of the flow of supply chain are described below.

Transportation

Transportation or shipment is necessary for an uninterrupted and seamless supply. The factors that have an impact on shipment are economic uncertainty and instability, varying fuel prices, customers' expectations, globalization, improvised technologies, changing transportation industry and labor laws.

The major elements that influence transportation should be considered, as it is completely dependent on these factors for order completion as well as for ensuring that all the flows work properly. The major factors are:

Long-term Decisions

Transportation managers should acknowledge the supply freight flow and accordingly design the network layout. Now, when we say long term decision, we mean that the transportation manager has to select what should be the primary mode of transportation.

The manager has to understand the product flows, volume, frequency, seasonality, physical features of products and special handling necessities, if any. In addition to this, the manager has to make decisions as to the extent of outsourcing to be done for each and every product. While considering all these factors, he should

carefully consider the fact that the networks need not be constant.

For example, in order to transport stock to regional cross dock facilities for sorting, packaging and brokering small loads to individual customers, stock destinations can be assembled through contract transportation providers.

Lane Operation Decisions

These functional decisions stress on daily freight operations. Here, the transportation managers work on real time information on products' requirements at different system nodes and must collaborate every move of the product that is both inbound and outbound shipping lanes so as to satisfy their services demands at the minimal possible cost.

Managers who make good decisions easily handle information and utilize the opportunities for their own profit and assure that the product is moved to them immediately, whenever it is demanded, that too in the right quantity. At the same time, they are saving cost on transportation also.

For example, a shipment has landed from a supplier who is based in New Jersey and in the same week, a product needs to be dispatched to New York as it becomes available for movement. If the manager is aware of this information in advance, he would prepare everything as per the demand and the products could be shipped out immediately.

Choice and Mode of Carrier

A very important decision to be made is to choose the mode of transportation. With the improvement in the means of transportation, modes of transport that were not available

in the traditional transportation modes in the past can be now be a preferred choice.

For example, rail container service may offer a package that is cost-efficient and effective as compared to a motor transport. While making a decision, the manager has to consider the service criteria that need to be met, like the delivery time, date special handling requirements, while also taking into consideration the element of cost, which would be an important factor.

Dock Level Operations

This involves the last level of decision-making. This comprises planning, routing and scheduling. **For example**, if a carriage is being loaded with different customers' orders, the function of the dock-level managers is to assure that the driver is informed of the most efficient route and that loads are placed in the order of the planned stops.

Warehousing

Warehousing plays a vital role in the supply chain process. In today's industry, the demands and expectations of the customers are undergoing a tremendous change. We want everything at our door step – that too with efficient price. We can say that the management of warehousing functions demands a distinct merging of engineering, IT, human resources and supply chain skills.

To neutralize the efficiency of inbound functions, it is ideal to accept materials in an immediately storable conveyance, like a pallet, case or box. For labeling the structure, tool selection and business process demand the types and quantities of orders that are processed. Further, the number of stock-keeping units (SKU's) in the distribution centers is a crucial consideration.

The Warehouse Management Systems (WMS) leads the products to their storage location where they should be stored. The required functionality for the completion and optimization of receiving, storing and shipping functions is then supplied.

Sourcing and Procurement

Sourcing and procurement are a vital part of the supply chain management. The company decides if it wants to perform all the exercises internally or if it desires to get it done by any other independent firm. This is commonly referred as the make vs buy decision, which we will be discussing in brief in another chapter.

Returns Management

Returns management can be defined as the management that invites the merger of challenges and opportunities for inbound logistics. A cost-effective reverse logistics program links the available supply of returns with the product information and demand for repairable items or re-captured materials. We have three pillars that support returns management processes. These are as follows:

Speed: It is a must to have quick and easy returns management and automate decisions regarding whether to produce return material authorizations (RMAs) and if so, how to process them. Basically, the tools of speed return processing include automated work flows, labels & attachments and user profiles.

Visibility: For improving the visibility and predictability, information needs to be captured initially in the process, ideally prior to delivering the return to the receiving dock. Most effective and easily implementable approaches for obtaining visibility are web-based portals, carrier integration and bar-coded identifiers.

Control: In case of returns management, synchronizing material movements is a common issue that needs to be handled. The producers need to be very cautious and pay close attention to receipts and reconciliation and update the stakeholders of impending quality issues. In this case, reconciliation activates visibility and control all over the enterprise. The key control points in this process are regulatory compliance, reconciliation and final disposition and quality assurance.

Software solutions can assist in speeding up the returns management by supporting user profiles and work flows that state supply chain partners and processes, by labeling and documentation that tracks the material along with the web-based portals and by exception-based reporting to deliver information for timely reconciliation. These characteristics, when executed with the three pillars mentioned above, support a reliable and predictable returns process to count value across the company.

Post-Sales Service

Now that the ordered shipment is over, what is the next step? The post sales service in supply chain tends to be an increasingly essential factor as businesses offer solution instead of products.

The post sales services comprise selling spare parts, installing upgrades, performing inspection, maintenance and repairs, offering training & education and consulting.

Presently, with the growing demands of the clients, a high volume of after sales service proves to be a profitable business. Here, the services are basically heterogeneous and the value-added services are different from those provided prior to sales service.

Supply Chain Management Strategy

Supply chain management (SCM) involves the movement of products and services from suppliers to distributors. SCM involves the flow of information and products between and among supply chain stages to maximize profitability.

The major functions involved in SCM are the procurement of raw materials, product development, marketing, operations, distribution, finance, and customer services. Customers are an integral part of SCM.

The objective of supply network or SCM is to maximize the overall value. Value is correlated to supply chain profitability. Here, profitability is the difference between the total revenue generated from the customer and the overall supply chain costs.

Strategies and designing of the supply chain include:

- Deciding on the supply chain structure and the activities each stage of the supply chain will perform
- Selecting a location and capacities of facility
- Deciding on the products that are to be made and the location where they need to be stored
- Choosing the modes of transportation and the source from where the information is to be collected
- Supply chain design decisions are long term projects and are expensive to reverse; so the manager must take into account the market

uncertainty.

7 Elements of Supply Chain Strategy

1.It's a multi-year road map - When people talk about strategy, they often say something like, "We want to reduce costs by 5%" or "we will do whatever it takes to achieve 95% service level". However, the Merriam-Webster dictionary defines strategy as,"A careful plan or method for achieving a particular goal, usually over a long period of time" So, the strategy should be a plan that takes many years to implement the project to achieve long-term business results, not just a set of "KPIs".

2. Start with customer needs - We believe many organizations incorporate customer satisfaction surveys into a customer service strategy formulation process which is a very good sign. Anyway, current customer needs may not be enough. We should try to anticipate future needs as well.

Another point about customer needs is that people in the supply chain function should try to "go see" customers so we can have a better understanding of the issues that might have an impact on demand/supply and the future of international business.

3. Include internal assessment SWOT analysis seems to be a standard method for an internal assessment. Anyway, you will be able to understand your own deficiencies much better if you do the business process analysis, even a simpler one.

4. Think about evolving technology Technology is the enabler for new processes and practices. From my observation, people read news related to supply chain technology so extensively on social media. My question is,

do we ever turn the news that we read everyday into something more meaningful like making a summary of the pros and cons of each technology?

5. Track domestic and foreign competitors

When your competitors offer some new products/services, these will eventually become the new customer's requirements. Then, we need to track competitors' movements so we can determine what customers might purchase in the future.

6. Consider megatrends - The technology outside the supply chain management area changes more rapidly, so we should track the technology outside the supply chain world too like lean manufacturing and six sigma.

7. It must be accepted across the company
A good strategy is not merely the action plans by each functional area, but a plan that is negotiated and agreed upon the team so people can take action and make decisions in the same direction

Value through the Supply Chain.

Maintaining a competitive advantage is a balance between providing great value for customers and doing it in such a way that your costs remain competitive. If you cut costs too much, you destroy your ability to service the customer. And service is the key ingredient behind customer retention. If your costs are too high, then your competition may gain an advantage in properly balancing costs with customer value. It should be noted that the value-chain encompasses all activities from design of products and services all the way through to the support of customers after they buy the product.

The buzzword for this whole process is called Supply Chain Management (SCM). The objective of SCM is to have all links in the chain working together, delivering products and services to customers when, where, and how they want it. At the same time, SCM must focus on minimizing costs and resources so that value is enhanced. Most value chains will consist of three links:

Distribution: Delivering products and services to customers.

Production: Converting resources into finished products and services according to the demands of the customer.

Resources: Acquiring the materials, people, and other resources to produce the required product or service.

Since this entire process is customer driven and since Distribution is closest to the customer, we start by looking at the Distribution link. Traditionally, distribution had several links, manufacturer to agent, agent to distributor, distributor to retailer, and retailer to customer. In today's global e-commerce world, it is quite common to see only two visible links: Manufacturer selling directly to the customer. By removing links, we cut down on lead times and reduce costs within the supply chain.

One useful tool for streamlining distribution links is the customer product map. A customer product grid or map will differentiate customers. For example, some customers prefer to buy direct while others prefer traditional distribution outlets. Customer maps also identify product mixes, geographic markets, seasonal patterns, and other relationships important to customer demand. Collecting information about customer demand is extremely important. The objective is to get the customer involved in driving Supply Chain Management (SCM).

Once distribution has been re-designed around the customer, the next step is to integrate production into distribution. You need a production process that is fast, flexible, and centered around the customer order. Gone are the days of producing standard products within standard cost accounting systems. The process is now customer driven with throughput accounting, no longer relying on production forecasts.

After production and distribution have been engineered to fit the marketplace, you can move to the Resource Area with SCM. Resources must be managed on a Just In Time basis, delivering materials and other resources only as they are needed. This will require that you educate suppliers as to your marketplace needs. Establishing strong relationships with everyone involved is critical. In some cases, you may need to phase-out certain suppliers in favor of the more networked, leading-edge suppliers you can fit with e-procurement applications. Additionally, it may be necessary to share costs with suppliers in order to retain relationships

Module II

Strategic Sourcing

Strategic Sourcing can be seen as a disciplined, systematic process for reducing the total costs of externally purchased materials, products and services while maintaining or improving levels of quality, service, and technology. It incorporates customer needs, organizational goals, company culture and philosophy, department bandwidth, and market conditions.

Sourcing used to be just another word for procurement some years back. As production becomes more streamlined, purchasing has taken on a more strategic role.

The evolution of Quality through modern management techniques like TQM, JIT, TOC, Six Sigma etc has underlined the importance of keeping the suppliers as partners.

Suppliers are important as most industries are affected by the following five actors.

- The need to control unit cost
- The need to reduce the total cost of acquisition;
- the purchaser's ability to respond to the customer's requirements on time
- An increased reliance on fewer suppliers;
- Dependence on suppliers to design and build entire sub assemblies and Sub systems

The Procurement function or Sourcing function encompasses the five key activities of

- Purchasing,
- Consumption management,
- Vendor selection,
- Contract negotiation, and
- Contract management.

While the market decides the price of the production in today's competitive business scenario, controlling the cost of operations decides the Profit of any organization.

The procurement (Sourcing) function is no longer to get the raw material at the lowest cost. But Quality, availability, and reliability are more significant now. Hence purchasing organizations are pursuing business in new ways, including strategic partnerships and sourcing sub assemblies rather than just single components. Sourcing can be categorized as Reactive, Tactical and Strategic Reactive Sourcing: As the name indicates, this type of sourcing occurs as reactive process. This is procurement when it becomes absolutely a necessary and is done to meet the immediate individual needs and when they arise. Since this is a transactional level sourcing and does not involve any real relationship with the supplier.

Tactical Sourcing: There are some business requirements that cannot be planned in advance. It takes a routine and sometimes reactive approach to purchasing using quick quote and order processes. It is a short term, transactional activity, practiced in small to medium size manufacturing organizations. To ensure the organization

has the right material at the right price and right time within the purchasing organization it is managed pro-actively. It does not strive to deeply understand vendors' core capabilities to support company's broader needs and is usually does not focus on requirements of the entire organization.

Strategic Sourcing is the core pivotal activity for purchasing and supply management professionals and requires extensive knowledge and competence. Strategic sourcing typically generates savings of 10-20% in direct materials and 15-25% in indirect goods and services. The Six "Rights" involved with Sourcing are

- Right Price: Material or service must have the right price.
- Right Quality: Material or service must be of right quality.
- Right Quantity: the supplier must supply in right quantity.
- Right On Time: Supplier must supply right on time.
- Right Source: the supply must be from the right source.
- Right Place: the supply must be from the right place

Success depends on selecting the right suppliers to participate in and then making themselves key players in those supply chains. Suppliers must be selected on the basis of quality, rather than solely on price. Without a measure of the quality being purchased, Price has no meaning. The focus should be on quality and life cycle costs rather than

price of material purchased. Minimizing the amount of incoming inspection and later corrective action would lead to minimization of overall costs.

The three key elements of Strategic sourcing are

Organized- Sourcing is part of an ongoing enterprise process covering all spending categories for materials and services. It is systematic and suppliers are invited to be part of the process

Systematic– Strategic Sourcing is fact-based and analytically rigorous.

Organization has an agreed-upon process for pursuing strategic sourcing .They are tracked in terms of cost savings and non-cost performance metrics as well. Further the process is regularly reviewed to incorporate new sourcing tools and approaches, new supplier market opportunities and developments in benchmarking and best practices.

Collaborative- Strategic sourcing teams are cross-functional, so suppliers must meet with a group that is armed with facts and looking at more than price. Supply managers collaborate internally, with department managers who use the materials and services purchases, with senior executives on key spend categories and externally, with key suppliers, incumbent or potential, on major new supply opportunities.

Characteristics of Strategic Sourcing: The main characteristics of Strategic sourcing are

- Disciplined buying practices and procedures
- A fair and equitable procurement process
- Respects the multiple goals of the public procurement environment.

- Puts total cost of ownership and best value in positions of primacy

Strategic Sourcing is important as it

- Helps to generate savings on the spending
- Helps to think creatively about Supplier Relationships
- Develops new thinking on Performance metrics
- Improves collaboration within the organization
- Lead to Quality improvement, improved team work etc

Strategic Sourcing answers to the most critical risk and cost management sourcing questions like

- Who do we buy from?
- What do we buy?
- How much should we buy from whom, when, and how?
- Is the supply base configured to future needs?
- How well are our suppliers performing to the goal?
- What is the best method to buy?

A strategic sourcing program can be segmented in many ways into a step-by-step approach. These steps must address the entire procurement value chain. A seven-step approach popularized by consulting firm A. T. Kearney (2004) is given below.

1. Assessment of a company's current spends(what is bought where?): An Internal Assessment is conducted to understand the current purchasing process. The assessment must be able to define the categories after validating the base information based on the identification of users, key constituencies and existing supplier relationships.

The purpose of Step 1 is to understand the organizations internal spend. It is one of the most time-consuming steps of the whole process so it is important to start well in advance of the target for completing a contract.

In this first Step Teams are formed to assess the current spend develop a comprehensive understanding of their category profile. The first task for the strategic sourcing team is to gather as much information as possible about what is bought, from whom, what are the product/service characteristics/ specifications, what is the current cost, and who and how the spend is managed internally.

They must understand market trends and economics that impact these purchases as well as the current negotiation leverage with suppliers. The methodology begins with a diagnostic of a company's spend with around 80% precision. A spend map by category is made by dividing the total corporate spend based on the supplier markets which is further categorized by the business units or locations, to the supplier level.

Spend analysis is the practice of reviewing procurement spend data by aggregating, classifying, and leveraging it. Such analysis can result in cost reduction, performance improvement, and greater compliance. Successful spend analysis relies on three critical factors:

- Access,
- Resources, and
- Powerful analytics.

A comprehensive spend analysis would more often reveal that among the various capabilities of an organization there are only few unique capabilities that are critical to revenue and offer competitive advantage. The other capabilities can be outsourced resulting in better utilization of the scarce resources. Spend analysis is the first step in the identification of possible savings opportunities and hence important. A complete spend assessment will provide visibility in company's needs, their total buying power and the degree to which they are leveraging that power.

One also need to interview key current users to develop a thorough understanding of their needs for the product, their view of the supplier's performance, and any enhancements that they would like to see in the product.

2. Conduct Market Assessment: A clear understanding of the external market is critical for determining the best sourcing strategy. This is basically an assessment of the Supply market (who offers what?) and consists of Analysis of the market dynamics/trends and identification of the Supplier universe.

One must understand the external supply market in which the supplier operates and the market pressures the supplier faces. There is a need to take a strategic and systematic approach to the market, rather than engaging with it on a tactical, short-term and project-by-project basis. Supplier can provide what is required only when he know exactly what is wanted. Hence supplier has to be given

detailed information about the product like specifications, quality requirements, and delivery cycle. Alert the market to potential requirement and take initial soundings on feasibility, capacity, capability, approach and level of interest.

These advertisements must clearly mention what the business needs is. The only potential suppliers can see what services are required and the organizational structure and skills and resources needed to deliver these. Suppliers can then assess whether they want and can do it.

3. Collect supplier information: In this step information as to where to buy what, while minimizing risk and costs is collected based on the market assessment. Shape the requirement and contractual options based on likely business priorities, market's views of risks, solutions and approaches and judge level of interest. Confirm realism of the requirement specification. This is followed by development of evaluation criteria by defining the performance metrics for the required materials. Based on these the spend data is verified.

4. Develop Sourcing strategy: The fourth step is the step for development of strategy. The specifications are fully developed in this step after identifying the sourcing levers. The end state scenarios are developed and potential suppliers are identified for each category.

Suppliers are to be selected using a competitive process whenever possible. The focus must be to create a supply management strategy that will govern the methods of securing a supply of the goods and services needed to continue a steady production. The number of suppliers for each category or item also needs to be decided. This is based on the availability, cost, the time needed, criticality etc of the item.

5. Evaluate the suppliers Bids: Analyzing suppliers and their bids thoroughly are very important. Failure to do so can lead to mistakes that could require costly and time-consuming rectification. The criteria to be used to assess both the supplier and the supplier's bid must be decided at the earliest stage itself. Evaluate potential service providers by establishing principles about contract flexibility and cooperative decision making that will govern the deal. Some of the criteria are

- Product and/or service suitability and compliance with the specification
- Delivery or completion date (compliance with programme stated in tender)
- Compliance with conditions of contract
- Price
- Whole-life cost or total cost of ownership
- Warranty
- Aesthetic characteristics
- Contribution to effective learning
- Improvements in management information

The use of scoring on a scale from 1 to 10 for various criteria is used to make the evaluation of suppliers easier. As some criteria would have greater significance depending on the items, suitable weightage may be given to arrive at the score.

6. Negotiate/select suppliers: Start with putting together a negotiations team consisting of the spokesperson, a senior

authority, a technical expert, a user, and an observer. Develop your negotiations strategy based on the information from RFPs and needs analysis, as well as knowledge of what is happening in the supplier market place and the suppliers' likely bargaining stance. Suppliers are selected in this step after negotiating with them. Factors that decide the selection of Suppliers generally are

- Faster delivery,
- Friendlier people,
- Easier credit terms,
- Better service, consistency (lack of variation) of product,
- Location,
- Branding and brand allegiance,
- Customizing,
- Various packaging,
- Lot sizes, etc.

According to Pyke and Johnson (2004) the following criteria are used in the selection of suppliers.

Production capabilities

- Capacity to produce the required items in the required quantities as per specification
- Availability of spare capacity

Technical capabilities

- Availability of the type of plant and equipment

required for producing the intended items in the quantity required.

- Availability of technically skilled and competent people capable of producing the intended item
- Availability of R & d facilities
- The quality control procedure of the supplier

Financial position of the supplier

- Type of ownership of the organization of the supplier
- Capital structure of the supplier
- Financial position and profitability for the last 5 years.

Other conditions

- Market standing of the supplier with regard to delivery commitments
- Market standing of the supplier with regard to quality commitments
- Working conditions of the supplier
- Industrial relations and power of the unions in the supplier organisation
- Supply of utilities like power Water etc.,

7. Implement Recommendations: The final step consists of the development of the category implementation plans,

communication plan and level of measurement and then implementing the plan.

Suppliers are rated based on the assessment of their performance for price, quality, delivery, and other important performance measures against another specific supplier or group of suppliers. It is a continuous process and involves objective appraisal of the supplier's performance, which can feed back to that supplier. For continuous improvement, the result of the implementation process is tracked and assessment is restarted. Supplier quality rating thus provides over a period of time.

The three key factors for a successful supply rating system are:

- An internal structure to implement and to sustain the rating program.
- A regular and formal review process and
- A standard measurement system for all the suppliers.

Suppliers can be rated based on the performance dimensions of quality, service, price and other factors as

- Conditional
- Approved
- Preferred and
- Strategic

The following measures or a combination of them are used for supplier rating.

- Product percentage non- conforming: ratio of the amount of defective items received to the total

number of items received

- Overall product quality: Points are assigned at incoming inspection and later phases of product application like production, supplier response to problems, customer complaints etc. These rating elements would have detailed criteria. Maximum points are allocated when there is no problem.
- Economic analysis: This analysis compares the suppliers on the total costs in monetary terms. This includes the quoted price of the item plus the costs associated with defect prevention, detection and correction.
- Composite plan: Apart from Quality, supplier performance depends on the basic categories of cost, Service, and delivery. It can have other categories and sub-categories. Weightage is given to these categories. Generally Quality is given the greatest weightage. Based on the above Performance Based Cost Ratio may be calculated for each of the supplier.

The final step in this process is supplier certification. Supplier Certification is a win-win customer-supplier alliance. The process of certifying begins with the supplier's performance. A certified supplier is one, who has proved to supply material of such quality that it is not necessary to perform routine testing on each lot received.

Supplier Certification is designed to identify the lowest cost, not price, supplier with a continuous improvement system to sustain long term cost reductions. It enables to develop a "shortlist" of qualified suppliers through a specific and proven set of criteria --a reduced supplier base that permits the most efficient use of the resources.

SUPPLY CHAIN COLLABORATION

Supply chain collaboration is defined as two or more autonomous firms working jointly to plan and execute supply chain operations. It can deliver substantial benefits and advantages to collaborators. It is known as a cooperative strategy when one or more companies or business unit's work together to create mutual benefits.

Types of Collaboration in Supply Chain Management

The types of collaboration in supply chain management pertain to the following:

Transaction Integration - Level one of collaboration in supply chain management pertains to the computerization of general business activities and transactions. In order to conduct these functions, you are able to utilize the Internet, EDI, or propriety software tools. Within this level, you and your supply team are able to exchange data such as sales orders, work orders, invoices, credit/debit notes, and more.

Supply Chain Management Information Sharing - Within this level, you are sharing information among various supply chain members using the Internet, EDI or proprietary software tools as well. Types of shared information within supply chain include production and transportation plans, bills of materials, production forecasts, orders, and more.

Strategic Collaboration - Once you have understood the other types of collaboration within supply chain management, you can come to strategic collaboration. Within this level, you and your supply chain partners will be collaborating on planning and the redesign process, which will usually enable you to share some risk and reward. You will be making decisions pertaining to

improving forecast accuracy, improving supply chain relationships, increasing profitability, and improving sales.

Benefits of Supply Chain Collaboration

- Lower inventory levels and higher inventory turns
- Lower transportation and warehousing costs
- Lower out-of-stock levels
- Shorter lead times
- Improved customer service metrics
- Visibility into customer demand and supplier performance
- Earlier and quicker decision-making

Inventory in SCM

Inventory management is the supervision of non capitalized assets -- or inventory -- and stock items. As a component of supply chain management, inventory management supervises the flow of goods from manufacturers to warehouses and from these facilities to point of sale. A key function of inventory management is to keep a detailed record of each new or returned product as it enters or leaves a warehouse or point of sale.

Organizations from small to large businesses can make use of inventory management to track their flow of goods. There are numerous inventory management techniques, and using the right one can lead to providing the correct goods at the correct amount, place and time.

Inventory control is a separate area of inventory management that is concerned with minimizing the total cost of inventory, while maximizing the ability to provide

customers with products in a timely manner. In some countries, the two terms are used synonymously.

Managing customer and vendor relationships is a critical aspect of managing supply chains. In many cases, the collaborative relationship concept has been considered the essence of supply chain management. However, a closer examination of supply chain relationships, particularly those involving product flows, reveals that the heart of these relationships is inventory movement and storage. Much of the activity involved in managing relationships is based on the purchase, transfer, or management of inventory. As such, inventory plays a critical role in supply chains because it is a salient focus of supply chains.

Perhaps the most fundamental role that inventory plays in supply chains is that of facilitating the balancing of demand and supply. To effectively manage the forward and reverse flows in the supply chain, firms have to deal with upstream supplier exchanges and downstream customer demands. This puts an organization in the position of trying to strike a balance between fulfilling the demands of customers, which is often difficult to forecast with precision or accuracy, and maintaining adequate supply of materials and goods. This balance is often achieved through inventory.

Inventory management techniques

Inventory management uses several methodologies to keep the right amount of goods on hand to fulfill customer demand and operate profitably. This task is particularly complex when organizations need to deal with thousands of stock-keeping units (SKUs) that can span multiple warehouses. The methodologies include:

- **Stock review**, which is the simplest inventory management methodology and is, generally, more appealing to smaller businesses. Stock review involves a regular analysis of stock on hand versus projected future needs. It primarily uses manual effort, although there can be automated stock review to define minimum stock levels that then enables regular inventory inspections and reordering of supplies to meet the minimum levels. Stock review can provide a measure of control over the inventory management process, but it can be labor-intensive and prone to errors.
- **Just-in-time (JIT) methodology**, in which products arrive as they are ordered by customers and is based on analyzing customer behavior. This approach involves researching buying patterns, seasonal demand and location-based factors that present an accurate picture of which goods are needed at certain times and places. The advantage of JIT is customer demand can be met without needing to keep large quantities of products on hand and in close to real time. However, the risks include misreading the market demand or having distribution problems with suppliers, which can lead to out-of-stock issues.
- **ABC analysis methodology**, which classifies inventory into three categories that represent the inventory values and cost significance of the goods. Category A represents high-value and low-quantity goods, category B represents moderate-value and moderate-quantity goods, and category C represents low-value and high-quantity goods. Each category can be managed separately by an inventory management system. It's important to know which items are the best sellers to keep enough buffer stock on hand. For example, more

expensive category A items may take longer to sell, but they may not need to be kept in large quantities. One of the advantages of ABC analysis is that it provides better control over high-value goods, but a disadvantage is that it can require a considerable amount of resources to continually analyze the inventory levels of all the categories.

- **Economic order quantity (EOQ) methodology**, in which a formula determines the optimal time to reorder inventory in a warehouse management system. The goal here is to identify the largest number of products to order at any given time. This, in turn, frees up money that would otherwise be tied up in excess inventory and minimizes costs.
- **Minimum order quantity (MOQ) methodology**, in which the smallest amount of product a supplier is willing to sell is determined. If a business can't purchase the minimum, the supplier won't sell it to them. This method benefits suppliers, enabling them to quickly get rid of inventory while weeding out bargain shoppers.
- **First in, first out (FIFO) methodology**, in which the oldest inventory is sold first to help keep inventory fresh. This is an especially important method for businesses dealing with perishable products that will spoil if they aren't sold within a specific time period. It also prevents items from becoming obsolete before a business has the chance to sell them. This typically means keeping older merchandise at the front of shelves and moving new items to the back.
- **Last in, first out (LIFO) methodology**, in which the newest inventory is typically recorded as sold first. This is a good practice when inflation is an issue and prices

are rising. Because the newest inventory has the highest cost of production, selling it before older inventory means lower profits and less taxable income. LIFO also means the lower cost of older products left on the shelves is what's reported as inventory. However, this is a difficult technique to put into practice, as older items that sit around have a chance of becoming obsolete or perishing.

- **Safety stock methodology**, in which a business sets aside inventory in case of an emergency. The safety stock approach also provides a signal that it's time to reorder merchandise before dipping into the safety stock. It's a good idea for businesses to work safety stock into their warehouse management strategy in case their supply chain is disrupted.

Module III

TRANSPORTATION SELECTION

Transportation plays a central role in seamless supply chain operations, moving inbound materials from supply sites to manufacturing facilities, repositioning inventory among different plants and distribution centers, and delivering finished products to customers. Benefits that should result from world-class operations at the points of supply, production, and customer locations will never be realized without the accompaniment of excellent transportation planning and execution. Having inventory positioned and available for delivery is not enough if it cannot be cost effectively delivered when and where needed.

Modes of transportation

There are six basic modes of transport that a company can choose from:

1. *Ship* which is very cost efficient but also the slowest mode of transport. It is limited to use between locations that are situated next to navigable waterways and facilities such as harbors and canals.
2. *Rail* which is also very cost efficient but can be slow. This mode is also restricted to use between locations that are served by rail lines.
3. *Pipelines* can be very efficient but are restricted to commodities that are liquids or gases such as water, oil, and natural gas.

4. *Trucks* are a relatively quick and very flexible mode of transport.

Trucks can go almost anywhere. The cost of this mode is prone to fluctuations though, as the cost of fuel fluctuates and the condition of roads varies.

5. *Airplanes* are a very fast mode of transport and are very responsive.

This is also the most expensive mode and it is somewhat limited by the availability of appropriate airport facilities.

6. *Electronic Transport* is the fastest mode of transport and it is very flexible and cost efficient. However, it can only be used for movement of certain types of products such as electric energy, data, and products composed of data such as music, pictures, and text.

Someday technology that allows us to convert matter to energy and back to matter again may completely rewrite the theory and practice of supply chain management.

Given these different modes of transportation and the location of the facilities in a supply chain, managers need to design routes and networks for moving products. A route is the path through which products move and networks are composed of the collection of the paths and facilities connected by those paths. As a general rule, the higher the value of a product (such as electronic components or pharmaceuticals), the more its transport network should emphasize responsiveness and the lower the value of a product (such as bulk commodities like grain or lumber), the more its network should emphasize efficiency.

Factors That Play a Key Role in Selecting Mode of Transport

1) Cost of Transport

When selecting the best and most suitable transport for exportation of products, the budget is the most important consideration. Costs vary based on the type and amount of goods needed to be transported. It is important to keep in mind that the cost of transport influences the cost of goods.

If heavy or bulky products are being transported over a long distance, inland, then rail transport is the most economical. Land transport, typically by trucks, is best suited for small amount of goods being transported over short distances, as it also saves packing and handling costs. The cheapest mode of transport is water transport, albeit the slowest too, but most suited for heavy or bulky goods that need to be transported over long distances, where time is not an important factor. Air transport is the best option for transportation of perishable, fragile and valuable goods, even though it is the most expensive.

It is important for importers and exporters to consider the overall cost of transportation, keeping the “hidden costs” such as insurance premiums and finance charges in mind.

2). Reliability and Regularity of Service

The reliability and regularity parameters of different transport modes, differ from each other. The urgency and speed by which the goods are to be delivered, influences the decision as to which mode of transport to use. All modes of transport, land, ocean and air, are affected by bad weather such as heavy rains, snow, fog and storms, which may cause delays.

3) Safety

Another crucial factor influencing the selection of a mode of transport is the safety and security of goods in transit. Land transport is more preferred to railway transport because the losses are less.

From the safety point of view, sea transport is the most risky, as water transport exposes the goods to the perils of sea, and the long duration of travel adds to the risk factors. Certain types of packaging also helps in safeguarding the goods in transit and are highly recommended, but they do influence costs as well.

Some goods also require special facilities such as refrigeration or special security measures that need to be taken into consideration when selecting a mode of transport.

4) Characteristics of goods

When selecting the mode of transport, the size and weight of goods play a crucial role. Land and air transport primarily cater to fragile and small shipments. Rail and sea transport are a more suitable option for heavy shipments.

How dangerous, fragile or high value the products are, also influences the selection of the transport mode. For breakable and fragile, high value products, air and land transport are the best option.

5) Budget

Transportation costs are add-on cost on the sale price of a product. The transport costs add to how much the goods can be sold for, to make a profit. Hence budget requirements need to be considered accordingly. The volume or weight of the goods plays a key role in determining which method of transport is the best value.

Also the type of good in terms of urgency of delivery, such as how perishable the goods are, and how quickly they need to arrive, influences the choice of mode of transport.

Typically, water transport is the most cost effective, and is mostly suitable when there is no hurry, and the goods are bulky or heavy, and need to be carried a long distance. Rail transport is relatively inexpensive for these types of goods. The advantage of road transport, in terms of cost is that there are savings to be made on the costs of handling and packaging. Air transport is one of the most expensive modes, but has the benefit of being fast, so is best suited for perishable and fragile goods.

6) Timescale

Air transport is the best option for long distances requiring urgent and speedy transport, to meet deadlines or because the goods are perishable or fragile. Motor transport is faster than rail transport for short distance deliveries. However, for longer haul journeys rail is faster and more economical. Water transport is often not suitable where time is a factor.

7) Flexibility

The most flexible mode of transport is Motor or road transport, as it is not constrained by factors such as flight times, shipping routes or pre-scheduled timetables. Motor transport can operate day and night, at personal convenience, to suit all time frames, and has the added advantage of door-to-door delivery.

DESIGN OPTIONS FOR A TRANSPORTATION NETWORK

The design of a transportation network affects the performance of a supply chain by establishing the infrastructure within which operational transportation decisions regarding scheduling and routing are made. A

well-designed transportation network allows a supply chain to achieve the desired degree of responsiveness at a low cost. We discuss a variety of design options for transportation networks and the strengths and weaknesses for each option in the context of a buyer with multiple locations sourcing from several suppliers. These design options may be implemented between any two stages of a supply chain.

DIRECT SHIPMENT NETWORK

With the direct shipment network option, the buyer structures his transportation network so that all shipments come directly from each supplier to each buyer location. With a direct shipment network, the routing of each shipment is specified and the supply chain manager only needs to decide on the quantity to ship and the mode of transportation to use. This decision involves a trade-off between transportation and inventory costs, as discussed later in the chapter.

The major advantage of a direct shipment transportation network is the elimination of intermediate warehouses and its simplicity of operation and coordination. The shipment decision is completely local, and the decision made for one shipment does not influence others. The transportation time from supplier to buyer location is short because each shipment goes direct.

A direct shipment network is justified if demand at buyer locations is large enough that optimal replenishment lot sizes are close to a TL from each supplier to each location.

With small buyer locations, however, a direct shipment network tends to have high costs. If a TL carrier is used for transportation, the high fixed cost of each truck

results in large lots moving from suppliers to each buyer location, resulting in high supply chain inventories. If a LTL carrier is used, the transportation cost and the delivery time increase, though inventories are lower. If package carriers are used, transportation costs are very high. With direct deliveries from each supplier, receiving costs are high because each supplier must make a separate delivery.

DIRECT SHIPPING WITH MILK RUNS

A *milk run* is a route on which a truck either delivers product from a single supplier to multiple retailers or goes from multiple suppliers to a single buyer location.

In direct shipping with milk runs, a supplier delivers directly to multiple buyer locations on a truck or a truck picks up deliveries destined for the same buyer location from many suppliers. When using this option, a supply chain manager has to decide on the routing of each milk run.

Direct shipping provides the benefit of eliminating intermediate warehouses, whereas milk runs lower transportation cost by consolidating shipments to multiple locations on a single truck. For example, the replenishment lot size for each buyer location may be small and require LTL shipping if sent directly. The use of milk runs allows deliveries to multiple locations to be consolidated on a single truck, resulting in better utilization of the truck and somewhat lower costs. Companies such as Frito-Lay that make direct store deliveries use milk runs to lower their transportation cost. If very frequent small deliveries are needed on a regular basis and either a set of suppliers or a set of retailers is in geographic proximity, the use of milk runs can significantly reduce transportation costs. For example, Toyota uses milk runs from suppliers to support its JIT manufacturing system in both Japan and the United

States. In Japan, Toyota has many assembly plants located close together and thus uses milk runs from a single supplier to many plants. In the United States, however, Toyota uses milk runs from many suppliers to each assembly plant.

ALL SHIPMENTS VIA CENTRAL DC

Under this option, suppliers do not send shipments directly to buyer locations. The buyer divides locations by geographic region and a DC is built for each region. Suppliers send their shipments to the DC and the DC then forwards appropriate shipments to each buyer location. The DC is an extra layer between suppliers and buyer locations and can play two different roles. One is to store inventory and the other is to serve as a transfer location.

In either case, the presence of DCs can help reduce supply chain costs when suppliers are located far from the buyer locations and transportation costs are high. The presence of a DC allows a supply chain to achieve economies of scale for inbound transportation to a point close to the final destination, because each supplier sends a large shipment to the DC that contains product for all locations the DC serves. Because DCs serve locations nearby, the outbound transportation cost is not very large. For example, W. W. Grainger has its suppliers ship products to one of nine DCs, which in turn replenish their almost 400 branches. If transportation economies require very large shipments on the inbound side, DCs hold inventory and send product to buyer locations in smaller replenishment lots.

For example, when Wal-Mart sources from an overseas supplier, the product is held in inventory at the DC because the lot size on the inbound side is much larger than the sum of the lot sizes for the stores served by the DC. If

replenishment lots for the buyer locations served by a DC are large enough to achieve economies of scale on inbound transportation, the DC does not need to hold inventory. In this case the DC can crossproduct arriving from many suppliers on inbound trucks by breaking each inbound shipment into smaller shipments that are then loaded onto trucks going to each buyer location. When a DC cross-docks product, each inbound truck contains product from a supplier for several buyer locations, whereas each outbound truck contains product for a buyer location from several suppliers. A major benefit of cross docking is that little inventory needs to be held and product flows faster in the supply chain. Cross-docking also saves on handling cost because product does not have to be moved into and out of storage. Successful cross-docking, however, does require a significant degree of coordination and synchronization between the incoming and outgoing shipments.

Cross-docking is appropriate for products with large, predictable demands and requires that DCs be set up such that economies of scale in transportation are achieved on both the inbound and outbound sides. Wal-Mart has used cross-docking successfully to decrease inventories in the supply chain without incurring excessive transportation costs. Wal-Mart builds many large stores in a geographic area supported by a DC. As a result, the total lot size to all stores from each supplier fills trucks on the inbound side to achieve economies of scale. On the outbound side, the sum of the lot sizes from all suppliers to each retail store fills up the truck to achieve economies of scale.

SHIPPING VIA DC USING MILK RUNS

As explained in milk runs can be used from a DC if lot sizes to be delivered to each buyer location are small. Milk runs reduce outbound transportation costs by

consolidating small shipments. For example, Seven-Eleven Japan cross-docks deliveries from its fresh-food suppliers at its DCs and sends out milk runs to the retail outlets because the total shipment to a store from all suppliers does not fill a truck. The use of cross-docking and milk runs allows Seven-Eleven Japan to lower its transportation cost while sending small replenishment lots to each store. The use of cross-docking with milk runs requires a significant degree of coordination and suitable routing and scheduling of milk runs.

The online grocer Peapod uses milk runs from DCs when making customer deliveries to help reduce transportation costs for small shipments to be delivered to homes.

OshKosh B'Gosh, a manufacturer of children's wear, has used this idea to virtually eliminate LTL shipments from its DC in Tennessee to retail stores.

TAILORED NETWORK

The tailored network option is a suitable combination of previous options that reduces the cost and improves responsiveness of the supply chain. Here transportation uses a combination of cross-docking, milk runs, and TL and LTL carriers, along with package carriers in some cases. The goal is to use the appropriate option in each situation. High demand products to high-demand retail outlets may be shipped directly, whereas low demand products or shipments to low-demand retail outlets are consolidated to and from the DC. The complexity of managing this transportation network is high because different shipping procedures are used for each product and retail outlet. Operating a tailored network requires significant investment in information infrastructure to facilitate the coordination. Such a network, however, allows for the

selective use of a shipment method to minimize the transportation as well as inventory costs.

TRADE-OFFS IN TRANSPORTATION DESIGN

All transportation decisions made by shippers in a supply chain network must take into account their impact on inventory costs, facility and processing costs, the cost of coordinating operations, as well as the level of responsiveness provided to customers.

For example, Dell's use of package carriers to deliver PCs to customers increases transportation cost but allows Dell to centralize its facilities and reduce inventory costs. If Dell wants to reduce its transportation costs, the company must either sacrifice responsiveness to customers or increase the number of facilities and resulting inventories to move closer to customers.

The cost of coordinating operations is generally hard to quantify. Shippers should evaluate different transportation options in terms of various costs as well as revenues and then rank them according to coordination complexity. A manager can then make the appropriate transportation decision. Managers must consider the following trade offs when making transportation decisions:

- Transportation and inventory cost trade-off
- Transportation cost and customer responsiveness trade-off

TRANSPORTATION AND INVENTORY COST TRADE-OFF

The trade-off between transportation and inventory costs is significant when designing a supply chain network. Two fundamental supply chain decisions involving this trade off are

- Choice of transportation mode
- Inventory aggregation

Choice of Transportation Mode

Selecting a transportation mode is both a planning and an operational decision in a supply chain. The decision regarding carriers with which a company contracts is a planning decision, whereas the choice of transportation mode for a particular shipment is an operational decision. For both decisions, a shipper must balance transportation and inventory costs. The mode of transportation that results in the lowest transportation cost does not necessarily lower total costs for a supply chain. Cheaper modes of transport typically have longer lead times and larger minimum shipment quantities, both of which result in higher levels of inventory in the supply chain. Modes that allow for shipping in small quantities lower inventory levels but tend to be more expensive.

Dell, for example, airfreights several of its components from Asia. This choice cannot be justified on the basis of transportation cost alone. It can only be justified because the use of a faster mode of transportation for shipping valuable components allows Dell to carry low levels of inventory.

Inventory Aggregation

Firms can significantly reduce the safety inventory they require by physically aggregating inventories in one location. Most e-businesses use this technique to gain advantage over firms with facilities in many locations. For example, Amazon.com has focused on decreasing its facility and inventory costs by holding inventory in a few warehouses, whereas booksellers such as Borders and

Barnes & Noble have to hold inventory in many retail stores.

Transportation cost, however, generally increases when inventory is aggregated. If inventories are highly disaggregated, some aggregation can also lower transportation costs. Beyond a point, however, aggregation of inventories raises total transportation costs. Consider a bookstore chain such as Borders. The inbound transportation cost to Borders is due to the replenishment of bookstores with new books. There is no outbound cost because customers transport their own books home. If Borders decides to close all its bookstores and sell only online, it will have to incur both inbound and outbound transportation costs. The inbound transportation cost to warehouses will be lower than to all bookstores. On the outbound side, however, transportation cost will increase significantly because the outbound shipment to each customer will be small and will require an expensive mode such as a package carrier. The total transportation cost will increase on aggregation because each book travels the same distance as when it was sold through a bookstore, except that a large fraction of the distance is on the outbound side using an expensive mode of transportation. As the degree of inventory aggregation increases, total transportation cost goes up. Thus, all firms planning inventory aggregation must consider the trade-offs among transportation, inventory, and facility costs when making this decision.

Inventory aggregation is a good idea when inventory and facility costs form a large fraction of a supply chain's total costs. Inventory aggregation is useful for products with a large value-to-weight ratio and for products with high demand uncertainty. For Example, inventory aggregation is very valuable for new products in the PC

industry, because PCs have a large value-to-weight ratio and demand for new products is uncertain.

Inventory aggregation is also a good idea if customer orders are large enough to ensure sufficient economies of scale on outbound transportation. When products have a low value-to-weight ratio and customer orders are small, however, inventory aggregation may hurt a supply chain's performance because of high transportation costs.

Compared to PCs, the value of inventory aggregation is smaller for best-selling books that have a lower value-to-weight ratio and more predictable demand.

TRADE-OFF BETWEEN TRANSPORTATION COST AND CUSTOMER RESPONSIVENESS

The transportation cost a supply chain incurs is closely linked to the degree of responsiveness the supply chain aims to provide. If a firm has high responsiveness and ships all orders within a day of receipt from the customer, it will have small outbound shipments resulting in a high transportation cost. If it decreases its responsiveness and aggregates orders over a longer time horizon before shipping them out, it will be able to exploit economies of scale and incur a lower transportation cost because of larger shipments. *Temporal aggregation* is the process of combining orders across time.

Temporal aggregation decreases a firm's responsiveness because of shipping delay but also decreases transportation costs because of economies of scale that result from larger shipments. Thus, a firm must consider the trade-off between responsiveness and transportation cost when designing its transportation network.

THE ROLE OF IT IN TRANSPORTATION

The complexity and scale of transportation makes it an excellent area within the supply chain for the use of IT systems. The use of software to determine transportation routes has been the most common IT application in transportation. This software takes the location of customers, shipment size, desired delivery times, information on the transportation infrastructure (such as distances between points), and vehicle capacity as inputs. These inputs are formulated into an optimization problem whose solution is a set of routings and a packing list for each vehicle that minimize costs while meeting delivery constraints.

Along with routing, vehicle packing software helps improve fleet utilization. By accounting for the size of the container and the size and sequence of each delivery, this software develops a plan to pack the vehicle efficiently while allowing for the great ease of unloading and/or loading along the route. Synchronization between the packing and routing software is important because how much is packed on a truck affects the routing, while the routing obviously affects what is packed on a truck.

IT also comes into play in the use of Global Positioning Systems (GPS) and electronic notification of impending arrivals. GPS systems monitor the real-time location of vehicles. This real-time information improves a firm's response to customer questions regarding deliveries. Electronic notifications are useful in a wide variety of situations, ranging from U.S. Customs receiving a ship's manifest days, or even weeks, before the ship docks in a U.S. port to a truck notifying a warehouse's receiving department that it expects to arrive 30 minutes early. Notification tools allow those receiving goods to be

prepared when the vehicle arrives and thus to perform all the loading, unloading, and corresponding administrative work in a more efficient manner.

The most common problems in the use of IT in transportation relate to cross enterprise collaboration and the narrow view taken by some transportation software. Collaboration across enterprises is crucial in transportation because this is a function that is often outsourced and does not directly involve either the shipper or the customer.

Thus, successful collaboration in transportation requires three or more firms to work together, making it much more difficult. Other problems arise because much of the transportation software is very focused on efficient routings. The software often overlooks other factors such as customer service and promised delivery times, which should constrain the route selected.

The players in this space include virtually all the major supply chain vendors, including the ERP firms and the supply chain-focused best-of-breed firms. There has also been a large amount of in-house development that has focused on the transportation management process.

Factors to Consider When Choosing a Distribution Channel

There are several channels available for the purpose of distribution of goods. Each channel has its own advantages and limitations and every company has to make difficult choice about channels of distribution. This decision about choice of a channel of distribution depends on several factors. A company has to consider all these factors and make an appropriate choice.

The following are the factors:

1. Market Related Factors:

Since the channels of distribution operate in the market. The market related factors are very important. There are several forces in the market which dictate the choice of channels of distribution.

The following are the market related factors to be considered:

a. Customers:

The ultimate purpose of any channel of distribution is to distribute the goods to the customers. Therefore the requirements and the nature of the customers should be considered while deciding the channel of distribution.

If the customers are widely scattered the channels must be in a position to reach them out effectively. This requires appropriate channels but if the customers are not widely scattered smaller channels would be sufficient. If the customers are very large in number such as individuals, very wide channels of distribution will be necessary, but if the customers are small in number and purchase in large quantities such as the industrial purchasers, small channels or even direct distribution will be sufficient.

b. Competition:

One has to consider the channels of distribution arranged by the competitors. This choice represents the wisdom and experience of the competitors. It also means that the competitors have been successful in using such channels over the long run. A company can adopt such channels of distribution if found suitable to itself. Unless there are compelling reasons, a company should not try to

change the pattern of distribution as compared with that of the competition.

c. Existing Channels of Distribution:

One has to make study of the existing channels of distribution. The functions performed by these channels, their strengths and weaknesses, their suitability and such other factors affect the choice of channels. Their relative advantages must also be studied.

2. Product Factors:

Since it is the product which is to be distributed, the product characteristics also have to be analyzed while choosing a channel of distribution. Different products are different in nature and this nature of the products requires different types of channels.

The following product factors have to be considered:

a. Perish ability:

If the products are highly perishable, the channel must be short or even direct marketing would be suitable. This is because long channels of distribution with a large number of intermediaries delay the distribution of goods. Products like milk, flowers etc. require very fast distribution.

b. Nature of the Product:

Consumer goods are purchased by a larger number of people, in smaller quantities and more frequently. Therefore such goods require longer channels of distribution which have a wide range. The presence of retailers is a must. Industrial goods on the other hand are purchased in larger quantities by a smaller number of purchasers and less frequently. Moreover the industrial goods purchaser is well

informed, knowledgeable and rational. Such goods require shorter channels of distribution.

c. Technicality:

Some products are highly technical in nature such as computer hardware and software, medical diagnostic equipments etc. Such goods require a high amount of technical support which can be provided only by the manufacturer. Therefore, such goods are best distributed by manufacturers' salesmen. Goods which do not require such technical support, for example-ready-to-wear garments can be distributed by longer channels of distribution.

d. Seasonality:

Some goods have a seasonal nature either in terms of production (agricultural goods), or consumption (woolen goods) and such goods require different types of distribution channels.

e. Variety Offered:

If a manufacturer has a wide range of goods, he can opt for direct distribution of the goods since a large number of products are available. If a manufacturer has very few products, he has to distribute them through long channels of distribution.

f. Unit Value:

Products of high unit value suit shorter channels of distribution or even direct marketing, but products of low unit value which are mass consumed require longer channels of distribution.

3. Company Factors:

A company has to look within and understand itself while choosing a channel of distribution. It has to understand its requirements, strengths and weaknesses.

The following specific factors have to be understood:

a. Company's Financial Strength:

A financially strong company can design its own channel of distribution because of its financial strength. It can negotiate with people and establish an altogether new channel of distribution. A company which is not financially strong has to settle down for existing channels of distribution because establishing a new channel of distribution requires huge amounts of money.

b. The Extent of Control Desired:

Control desired in this context means the ability of the company to exercise control over the channels of distribution in matters like resale price maintenance, territory restrictions etc. Longer the channel, lesser will be the control.

c. Reputation of the Company:

A well-established company with a strong reputation will find it easy to have longer channels of distribution. This is because channel intermediaries are generally willing and enthusiastic to be associated with strong companies.

d. Company's Marketing Policies:

Every company will have policies regarding marketing and these policies will also lay down norms relating to channels of distribution and these policies will

also have a strong influence on the choice of channels of distribution.

e. Past Experience:

An established company will already have well established channels of distribution. The company will also have experience in matters of dealing with such channels of distribution. A company should consider such past experience while deciding the channels of distribution.

4. Channel Related Factors:

The channels of distribution choosing should be appropriate from the view point of the company. These channels must be examined and then a proper choice must be made.

The following factors of the channel must be considered:

a. The Ability of the Channels:

Well established and strong channels have the ability to distribute goods effectively over a wide area. They can promote and sell even unknown products. Newly established channels of distribution however cannot do these. Therefore a company has to consider the ability of the channels before deciding on the channels of distribution.

b. The Financial Strength of the Channels:

Financially strong channels of distribution can distribute the goods well and also finance the manufacturers directly or indirectly. They can lift the goods from the manufacturers by paying cash immediately which indirectly amounts to financing the manufacturers. Therefore, such financial ability is also a factor that must be considered by a company regarding choice of channel of distribution.

c. Ability to Provide after Sales Service:

Some products require a long term after sales service. In such a case it should be decided as to who has to provide the after sales service whether the manufacturer or a member of the channel of distribution. In such a case a company has to look into the ability of the channels of distribution to provide effective after sales service in a sustained manner.

5. Environmental Factors:

A company's channel choice depends on certain environmental factors. Environment in this context means the environment within which the company, the channels of distribution, the customers etc. are present.

The following are certain environmental factors which must be considered while deciding channels of distribution:

a. Economic Situation:

The prevailing economic situation in the country affects all the economic activities. Therefore, a company has to be aware of the prevailing economic conditions. During an economic boom, the sales of all the products will naturally good and channels of distribution will be more than willing to take up distribution of products.

During a recessionary period, the general sales come down as a result of which channels of distribution become reluctant to take up distribution. These are the factors to be considered by a company while deciding on a channel of distribution.

b. Legal Factors:

A company is free to decide about its channels of distribution as long as its activities are legal. There are

certain legal factors however which must be considered while deciding channels of distribution and arrangements with them. Certain types of arrangements with the channels of distribution in the form of sole distributorship and in the cases of certain essential commodities may be objectionable under law. Therefore, such legal factors are to be considered.

c. Fiscal Structure:

Fiscal structure in this context refers to certain indirect taxes levied by the state governments on products. There is no uniformity in this matter and clarity is absent in certain cases. Therefore, such matters must also be considered while deciding on, channels of distribution.

Third Party Logistics (3PL)

Third party logistics or 3PL is a service that enables enterprises to outsource or make use of third -party businesses to carry out tasks to completion in its supply chain.

- 3PL providers are companies that offer comprehensive and even one-stop solutions for enterprise supply chains in packaging, warehousing, transportation and order fulfilment.
- 3PLs can identify and fill the gaps in your supply chain.
- Their expertise, best practices and technologies can integrate into your business processes to solve the most complex challenges in logistics management and make your supply chain a more agile, responsive function — a necessity in today's uncertain market environment.

Working with a 3PL provider is a partnership and must be treated as such rather than as a transactional relationship.

Benefits of 3PL in supply chain management:

- **Drive Cost Savings**

Third Party Logistics firms specialize in logistics and thus will have a more extensive network than your company's supply chain function. They will likely have exclusive relationships within the logistics sector, greater influence during negotiations, and will also be able to offer greater volume discounts to clients. All of this can minimize overhead costs.

By partnering with a 3PL supply chain management firm, you can also save on making huge infrastructure investments as it can provide transportation, warehouse space, staff and tracking technology, among other things.

- **Get Access to Expertise and Experience**

In today's complex global market scenario, it is tough to anticipate and accommodate internal expertise in all the capacities and regions required. A 3PL service provider will have knowledge and experience in matters such as transport documentation, import and export, international compliance and economic regulations, for instance. Businesses looking to expand into international markets can benefit from the logistics support and know-how that their partner can provide, thereby reducing costly delays, cutting down the cycle time, and making the entry into a new region smoother.

- **Focus on Core Competencies**

Outsourcing logistics will give your organization the leeway to focus on its core competencies instead of getting involved in the management of non-core but critical functions. Your business can enjoy the benefits of logistical expertise without deploying internal resources.

- **Gain Flexibility and Scalability**

The advantage of third-party logistics in supply chain management is that it offers enterprises the flexibility and scalability to utilize supply and distribution resources based on current business needs. Thus, when sales are down, there are no redundant investments and utilized resources, and when there's a surge in demand, enterprises can upscale.

- **Enable Business Growth and Market Expansion**

The role of third-party logistics in supply chain management is to enable business growth by giving companies access to markets where they don't have an established presence. Being able to manage inventory in a new market without having to spend money on warehousing, equipment, and labor can save money, as well as the effort of learning the logistical nuances of a new market.

Improve Customer Satisfaction

All of the aforesaid 3PL benefits will lead to improved services and response time, timely deliveries and greater brand reliability. This translates to satisfied customers — the fundamental goal of all businesses.

INDIAN TRANSPORT INFRASTRUCTURE

India's transport system is one of the largest in the world serving the land mass of 3.3 million square km and a population of over one billion. The network for the services comprise mainly of roads, railways and air services. The 6000 km coast line has 12 major and over 150 minor ports. While the ministry of Railways is responsible for running of the rail system, the highways are the joint responsibility of the Central Ministry of Road Transport and Highways (for National Highways) and the State Governments (State

Highways, District and Rural roads). The Ministry of Civil Aviation looks after Airports and the Ministry of Urban Development responsible for Urban Transport.

The growth in the network of transport and the demand for transport services in the last 50 years has been quite significant. Currently, about 800 billion tonne kms of freight and 2300 billion passenger kms¹ is handled by the transport system, with the total transport demand growing at around 10 percent a year in the last 10 years. The modal growth rates have varied with road transport and air services growing at a much higher rate than the railways.

Railways which had a market share of 88% of freight and 68% of the passenger business in land transport in 1950-51 has come down to about 40% of freight and about 20% of passenger market share in 1999-2000 while road transport has moved from 12% to 60% in freight and from 32% to 80% in the passenger business in the same period.

Transport Infrastructure in India has a share of 6.4 per cent in GDP and its demand has been accelerating over the years.

Roads

An important component of transport infrastructure is road transport. The road length per square kilometer of land area is known as density of roads. The road density has been increasing over the years in India. For instance road density in India stood at 714 per sq km of land area in 1991, increased to 1008 per sq km during 1991 and stood at 1171 per sq. km in 2007. In-fact, among the BRIC economies, road density in India is quite high. In Asia, only two countries namely Singapore and Japan have a better road density than India. However, most highways in

India are narrow and congested with poor surface quality, and 40 percent of India's villages do not have access to all-weather roads.

Another important element of roads is the paved roads as a percentage of total roads. Paved roads are those surfaced with crushed stone (macadam) and hydrocarbon binder orbituminized agents, with concrete, or with cobblestones, as a percentage of all the country's roads, measured in length.

However, in several other countries like Russia, Korea, Malaysia and Japan paved, roads as a percentage of the total roads are quite high as compared to India. In Singapore, paved roads as a percentage of total roads are 100 per cent followed by Malaysia and Russia where the percentage is more than 80 per cent. Further, goods transported by road are the volume of goods transported by road vehicles, measured in millions of metric tons times kilometers traveled. Goods transported are the maximum in China followed by India. But over the years, goods transported using road network is growing much faster in India along with China. This fast increase in traffic in India creates much pressure on the existing roads and thereby there is a demand for new roads. Another important parameter highlighting the importance of road infrastructure is the energy consumption of roadways. Road sector energy consumption is the total energy used in the road sector including petroleum products, natural gas, electricity, and combustible renewable and waste. This percentage is high for countries like Malaysia, Korea and Japan and low for countries like China and India.

Railways

Rail density in India is the third highest in Asia after Japan and South Korea. Railway density in China stood at 6.51 per cent in 2008 as compared to India's 21.20 percent. The railways have been recording consistent growth rates in the freight and passenger traffic. Goods transported by Indian railways are the second highest in Asia after China. Similarly, number of passengers carried by Indian railways is also on par with China indicating that railways are an important source of transportation to the majority of populations. Overall, the growth of passenger and freight traffic in case of India between 1991 to 2008 is substantial while rail density is at same level. Over the years growing demand has led to introduction of hundreds of new trains on the same tracks with almost same supporting logistics. This has resulted in a compromise on safety, hygiene and also overall efficiency of Indian railways.

Airways

Aviation Industry in India is one of the fastest growing aviation industries in the world. With the liberalization of the Indian aviation sector, aviation industry in India has undergone a rapid transformation but much more is desired in this important mode of transport infrastructure. Freight traffic of airways in India increased to 1233 million ton kms in 2008 as compared to only 493 million ton kms in 1991. However, countries like China, Brazil, Malaysia, Russia, Japan, Korea and Singapore have a better freight million ton-kms in comparison with India. Similarly, the number of passenger carried by airways in India was 49.9 million in 2008 *vis-à-vis* 191 million in China, 58.8 million in Brazil and 97.02million in Japan. This also shows that

railways and road transport in the country are more popular for a majority of Indian population who belong to poor and lower middle class where as growing middle income population has shifted to using air transport which has created both demand and supply of air services in the country. India is also ranked way below other countries, particularly China, Singapore and Japan in registered carrier departures world wide.

Ports

India has an extensive coastline of 7517 km, excluding the Andaman & Nicobar Islands. Indian ports handle around 95% of the total volume of country's external trade and about 70 per cent in terms of value. India has 12 major ports and 200 non major ports (minor and intermediate ports) spread across nine (coastal) maritime states. Total cargo volume in million tones handled by Indian ports increased from 156 million tones in 1991 to nearly 834million tones in 2008. In fact, the cargo handled by Indian ports is the second highest in Asia after China. The growth of cargo volume over the years is one of highest among the countries and there is growing pressure on the existing port facilities resulting in high pre-berthing and shipment turn around time. Similarly container traffic has also increased considerably since 1991 to reach 6623TEU's (twenty foot equivalent units) in 2008. However, port infrastructure in India is not up to international standards and India is ranked at a poor 105 in a set of countries handling port traffic.

The modernization of Indian ports needs huge resources and time bound efficient management.

According to the Global competitiveness Index published by the World Economic Forum, India is ranked

49 on the quality assessment of a range of indicators. Specifically, even with regard to infrastructure, India is ranked 76 with a score of 3.46 indicating that there is immense scope for further development of infrastructure facilities in India including transport infrastructure to catch up with other economies.

ELECTRONIC DATA INTERCHANGE (EDI)

Electronic data interchange (EDI) is an inter-organizational information system that is intended to facilitate computer-to-computer exchange of data among multiple business trading partners in standard, machine readable formats through a web of business-to-business communication network. EDI is useful for efficient and effective supply chain management, because EDI can help reduce lead time, save documentation processing cost, eliminate procurement errors, clarify inventory status information, and enhance strategic alliances throughout the supply chain. In a broader perspective, EDI plays three important roles in supply chain management. These roles are classified as:

- (1) Electronic integration;
- (2) Information diffusion and sharing; and
- 3) Electronic marketplaces.

EDI Challenges & Priorities

EDI is critical for many respondents, but there are challenges associated with implementing, using and maintaining the technology. We asked an open response question about the biggest challenges organizations had with EDI. The most common responses included:

- Difficulties finding qualified staff to manage EDI systems
- Time, particularly time to get set up with new EDI partners

-Partner requirements, both the lack of standardization between them and the frequency of updates.

-Cost of implementing and maintaining EDI

EDI helps organizations meet a range of objectives for supply chain management. In order, the main priorities organizations have for EDI are:

1: Reducing processing time

2: Improving productivity

3: Reducing errors

4: Reducing charge backs

5: Improving inventory and warehouse management

6: Streamlining their supply chain

7: Improving trading partner relationships

8: Better managing high volumes of data

SUPPLY CHAIN MANAGEMENT IN E-COMMERCE

Business efficiency at all operational levels is critical for the success of e-commerce. Supply chain management is an essential component of e-commerce. Supply chain management in e-commerce focuses on procurement of raw material, manufacturing, and distribution of the right product at the right time. It includes managing supply and demand, warehousing, inventory tracking, order entry, order management, distribution and delivery to the customer.

Following are the major advantages of supply chain management in e Commerce:

Visibility across the entire network

Supply chain management brings transparency in the network and helps oversee the status of all the activities happening across supply, production, warehousing, and distribution. This ensures a more comprehensive tracking and management of all processes from procuring to shipping of finished goods to the end consumer.

Enhanced Customer Relationships

Effective supply chain management ensures on-time deliveries, which makes a direct impact on strengthening customer relationships. Furthermore, SCM aids the brands to keep an eye on customer requirements. It makes sure that the business is attuned to the changes in the demands of the products and services. An e Commerce integrated supply chain helps businesses get feedback and requirements about their products directly from the consumers.

Cost reduction

One of the principal reasons due to which the customers invest their time and money in e Commerce is reduced costs. Probably, there are a lot of areas where business invests more than required. Some of such areas could definitely be streamlined. It's worth taking a look at your supply chain to recognize areas where the costs could be cut down.

e Commerce based SCM can significantly reduce costs in the supply, production, and distribution processes, which, in turn, leads to higher profits.

Minimized Delivery Delays

Delayed shipment from vendors, holdups during production, and logistic errors in distribution channels massively impact brand reputation in the market. With an

effective SCM, all activities can be coordinated and executed from top to bottom. Minimizing delivery delays can prevent losing out on business and streamline your image and relationships with the customer.

Scaling omni channel practices

A well-structured supply chain management in e Commerce facilitates omni channel engagement that ultimately leads to increased customer engagement on all major platforms.

e Commerce businesses in an omni channel environment pretty much conclude the new shopping touch points of the majority of consumers today. It is incredibly crucial for brands to consider customer convenience and expectations first. And, it starts from the very first step of the supply chain. An adequately managed e Commerce supply chain assists brands with an omni channel scaling of their businesses and expands their selling horizons with content customers spread across channels.

E-PROCUREMENT

In general, online procurement solutions can be defined as a process of purchasing or selling of goods and services through various electronic methods, through using the Internet. This is an alternative method for the manual procurement process. However, an electronic procurement solution possesses the potential to curb irregularities in the purchasing of business products and services as well as helping to reduce any unnecessary costs.

Benefits of E-Procurement in Businesses

The online procurement automation software offers numerous benefits for businesses of all sizes and shapes:

1. **Reduce Manual Errors:** The total procure to pay process becomes automated with the implementation of electronic procurement software solutions. This can help businesses to minimize the occurrence of human errors and increase the business efficiency.
2. **Enhance the Business Productivity:** Leveraging the solutions of an electronic procurement system is an added advantage for businesses. It can help businesses to streamline the purchasing and tendering processes, keep a track of the total purchases within the business organization, improve the manufacturing cycles, and automate the supply chain processes with automatic notifications, as well as improve the employee work productivity.
3. **Eliminate Unnecessary Paper Works:** The usage of electronic procurement tools enable businesses to store the business data in an electronic format, eliminating the paper work and simplifying the process of purchasing.
4. **Minimize the Business Costs:** The usage of e-procurement systems can enable businesses to gain control and visibility over the businesses spending, eliminate the unnecessary paper work, effectively manage the day-to-day business transactions, identify the procurement trends and metrics, and discover the various innovative methods to reduce business spending.
5. **Improve the Business Transaction Speed:** The services offered by electronic procurement can allow businesses to support multiple tasks, simplify the online purchasing process, establish meaningful relationships with suppliers, minimize the number of manufacturing cycles, and reduce the manual data entry. Moreover, the solution effectively manages day-to-day purchases, enhances visibility into

business transaction speed, and helps you gain competitive advantage over other businesses.

6. **Gain Visibility into Business Spending:** The centralized tracking system present in electronic procurement software can effectively track the business transactions as well as provide a detailed overview report on requisition, items purchased, supply chain processes, purchase orders, inventories, receipt and invoice management etc. All these reports can help businesses to gain enhanced visibility into the day-to-day business transaction process.
7. **Faster Approval Process:** Today, integrations are ruling the world of business. In market, you can find a few procurement softwares' that can offer fully integrated services. For example, in order to have a flexible approval process, businesses can integrate the electronic procurement system with an e-signature solution for the faster approvals, as well as to accomplish the job in a much easier format

BARCODES, QR CODES, & RFID IN SUPPLY CHAIN MANAGEMENT

Barcodes, QR Codes, and RFID tags play an important part in keeping the supply chain functional. They help companies improve the speed and accuracy of their processes by tracking inventory.

In recent years, [barcodes](#) have infiltrated the supply chain in rapid form, creating new opportunities for tracking and information gathering. From the simplest of goals – to see when a package will arrive at the consumer's doorstep – to more complex efforts such as tracking temperatures in transit, barcodes provide incredible information at a moment's notice. Consider the benefits bar codes bring to the process in supply chain management:

1. Tracking & Inventory Management
2. Enhancing Accuracy
3. Improving Transparency in the Process
4. Cost Effectiveness
5. Advantages of Bar codes over RFID Tags

1: Tracking & Inventory Management

Manufacturers and distributors can expedite the fulfillment process using [barcode scanning](#). A single bar code provides instant access to a large amount of data, including how a box should be packed or where on the warehouse floor it should be routed. The same bar code connects workers and truck drivers to managing paperwork, tracking when and where, and managing inventory more efficiently.

2: Enhancing Accuracy

The cost of a mistake is a big factor for all companies. With [barcode scanning](#), redundant information and error corrections can be made in real-time. Even if a package suffers damage or a label is destroyed. Solutions are available to use even limited data to complete transactions.

3: Improving Transparency in the Process

With granular visibility into the supply chain, there's more transparency to meet regulatory requirements. This is an increasing concern for many companies, especially those in the food industry who have to meet the demands of the FDA's Food Safety Modernization Act. Tracing products becomes more efficient and possible using bar code scanning technology. With more detail, companies can use

this information to manage recalls quickly, potentially saving lives in the process.

4: Cost Effectiveness

Companies need to strive to maintain a bottom line. Barcodes are one of the simplest and most cost-effective solutions for doing this. Though scanners used to be significantly more expensive in the past, today, they are far more affordable overall – and they offer more advanced features as well. A barcode label itself costs just a few cents.

Another key to consider is the savings they bring. [Barcode scanning technology](#) eliminates the costly need to manage labor costs. These systems are automated, requiring little to no human support. At the same time, labor costs continue to rise in the industry, driving up costs for human-managed supply chains.

5: Advantages of Bar codes over RFID Tags

Can RFID tags provide the same benefits as bar codes? Not necessarily. They require a significantly large extra investment. That's money companies often do not want (or cannot) spend. At the same time, RFID tags are not as durable as bar codes. That is, they can easily be damaged in transit, making them useless in providing rapid information when needed.

RFID Technology & its Applications

RFID stands for 'Radio-frequency identification' where digital data is encoded in RFID tags or smart labels which is captured by a reader. It is similar to bar coding where data from a tag is captured by a device. RFID belongs to a group of technology which includes automatic identification and data capture. Such technology

automatically identifies objects, collects data related to them and enters it directly in the computer with little or no human intervention.

Components of RFID system

It consists of three components – an RFID tag or label, a reader and an antenna. The tag contains a circuit and an antenna to transmit data to the reader. The reader in turn converts the waves which have been transmitted to a usable form of data. The information is then transferred to a host computer system, where data can be stored and analyzed.

Applications of RFID technology

1. **Logistics and supply chain** – With the visibility provided by RFID in the supply chain, It can help increase efficiency, reduce errors and improve on quality.
2. **Inventory tracking** – Tracking items through the supply chain to the point of sale can be done in a matter of minutes with the help of a RFID reader.
3. **Tracking attendees** – Managing a conference, with an RFID based solution will help eliminate the need for long registration lines.
4. **RFID based access** – In certain areas an increased level of security is required. RFID control tags can restrict access to only those people which have been pre-approved.
5. **Real time location system** – With this system, you can track the real time location of employees and assets. It helps gauge the effectiveness of a floor based plan as well as track the location of resources.

Benefits of RFID system

- Ease of payment – No need to carry cash for the toll transactions, saves time.
- Near non-stop movement of vehicles leading to lower fuel cost.
- Other benefits are:
 - Environmental benefit : reduced air pollution, reduced use of paper.
 - Social benefit : reduced toll payment hassles ,analytics for better highway management.
 - Economic benefit : reduced effort in management at toll plaza, reduced effort in monitoring centrally.

Module IV

LOGISTICS MANAGEMENT

As far back as history records, the goods that people wanted were not always produced where they wanted to consume them, or these goods were not accessible when people wanted to consume them. Food and other commodities were widely dispersed and were only available in abundance at certain times of the year. Early peoples had the choice of consuming goods at their immediate location or moving the goods to a preferred site and storing them for later use. However, because no well developed transportation and storage systems yet existed, the movement of goods was limited to what an individual could personally move, and storage of perishable commodities was possible for only a short time.

This limited movement-storage system generally constrained people to live close to the sources of production and to consume a rather narrow range of goods. Even today, in some areas of the world consumption and production take place only within a very limited geographic region. Striking examples can still be observed in the developing nations of Asia, South America, Australia, and Africa, where some of the population live in small, self-sufficient villages, and most of the goods needed by the residents are produced or acquired in the immediate vicinity. Few goods are imported from other areas. Therefore, production efficiency and the economic standard of living are generally low. In this type of economy, a well-developed and inexpensive logistics system would encourage an exchange of goods with other producing areas of the country, or even the world.

As logistics systems improved, consumption and production began to separate geographically. Regions would specialize in those commodities that could be produced most efficiently. Excess production could be shipped economically to other producing (or consuming) areas, and needed goods not produced locally were imported. This exchange process follows the *principle of comparative advantage*. This same principle, when applied to world markets, helps to explain the high level of international trade that takes place today. Efficient logistics systems allow world businesses to take advantage of the fact that lands, and the people who occupy them, are not equally productive. Logistics is the very essence of trade. It contributes to a higher economic standard of living for us all. To the individual firm operating in a high-level economy, good management of logistics activities is vital. Markets are often national or international in scope, whereas production may be concentrated at relatively few points. Logistics activities provide the bridge between production and market locations that are separated by time and distance. Effective management of these activities is the major concern of this Program.

MEANING AND DEFINITION OF LOGISTIC MANAGEMENT

The benefits of co-ordinated logistics management appeared around 1961, in part explaining why a generally accepted definition of business logistics is still emerging. Therefore, it is worthwhile to explore several definitions for the scope and content of the subject.

A dictionary definition of the term *logistics* is:

“The branch of military science having to do with procuring, maintaining, and transporting material, personnel, and facilities.”

This definition puts logistics into a military context. To the extent that business objectives and activities differ from those of the military, this definition does not capture the essence of business logistics management. A better representation of the field may be effected in the definition promulgated by the Council of Logistics Management (CLM), a professional organization of logistics managers, educators, and practitioners formed in 1962 for the purposes of continuing education and fostering the interchange of ideas. Its definition:

“Logistics is that part of the supply chain process that plans, implements, and controls the efficient, effective flow and storage of goods, services, and related information from the point of origin to the point of consumption in order to meet customers’ requirements.”

This is an excellent definition, conveying the idea that product flows are to be managed from the point where they exist as raw materials to the point where they are finally discarded. Logistics is also concerned with the flow of services as well as physical goods, an area of growing opportunity for improvement.

It also suggests that logistics is a **process**, meaning that it includes all the activities that have an impact on making goods and services available to customers when and where they wish to acquire them. However, the definition implies that logistics is part of the supply chain process, not the entire process.

Although early definitions such as physical distribution, materials management, industrial logistics and channel management - all terms used to describe logistics – have promoted this broad scope for logistics, there was little attempt to implement logistics beyond a company’s own enterprise boundaries, or even beyond its own internal

logistics function. Now, retail firms are showing success in sharing information with suppliers, who in turn agree to maintain and manage inventories on retailers' shelves. Channel inventories and product stock outs are lower. Manufacturing firms operating under just-in-time production scheduling build relationships with suppliers for the benefit of both companies by reducing inventories.

BUSINESS LOGISTICS

Business logistics is a relatively new field of integrated management study in comparison with the traditional fields of finance, marketing, and production. As previously noted, logistics activities have been carried out by individuals for many years. Businesses also have continually engaged in move store (transportation-inventory) activities. The newness of the field results from the concept of **co-ordinated management** of the related activities, rather than the historical practice of managing them separately, and the concept that logistics adds value to products or services that are essential to customer satisfaction and sales. Although co-ordinated logistics management has not been generally practiced until recently, the idea of co-ordinated management can be traced back to at least 1844. In the writings of Jules Dupuit, a French engineer, the idea of trading one cost for another (transportation costs for inventory costs) was evident in the selection between road and water transport:

“The fact is that carriage by road being quicker, more reliable and less subject to loss or damage; it possesses advantage to which businessmen often attach a considerable value.

However, it may well be that a saving induces the merchant to use a canal; he can buy warehouses and increase his floating capital in order to have a sufficient

supply of goods on hand to protect himself against slowness and irregularity of the canal, and if all told the saving in transport gives him a cost advantage, he will decide in favor of the new route.”

ELEMENTS OF LOGISTICS MANAGEMENT

Logistics management consists of eight elements called wings of logistics. These are discussed in a nutshell below.

1. Customer Order Processing

Flow of Actions

1. Filling up the order form
2. Deciding the specifications of the product
3. Deciding the quality check list of the product
4. Deciding the delivery schedule
5. Deciding the location of delivery

Important Factors

1. Cost of order processing
2. Whether the company is capable of producing a component
3. Detailed list of specifications

Techniques

1. Electronic data Interchange (EDI)
2. E-ERP or CPFR
3. Web portal

2. Location Analysis

Flow of Actions

1. Cost of transportation of raw materials and finished goods

- Proximity to suppliers
- Proximity to customers

2. Availability and type of land

3. Availability of secondary resources

4. Availability of desired manpower at affordable cost

5. Communal harmony

6. Governmental regulation and taxation

Important Factors

1. Cost of operations as a percentage of sales

2. Shelf life of product

3. Inventory Control

Flow of Actions

1. on hand inventory analysis

2. Communicating the quantity, quality and timing of material with the supply points.

3. Getting the material of right quality, quantity and at right time

Important Factors

1. Inventory control at planning stage

2. Lead time

3. Cost vs. importance of raw material

Techniques

1. DRP and replenishment order control

2. Fixed order interval system
3. Economic order quantity with ROP system
4. Selective inventory control (ABC, VED, FSN analysis etc.)
5. Order forecasting using statistical tools

4. Material Handling

Flow of Actions

1. Type of material (Business significance like raw material, finished goods etc.)
2. Material handling requirements of the material (Fragile, inflammable)
3. Cost ratio of material handling to material cost.
4. Material default location, identification and traceability

Important Factors

1. Material breakage
2. Pilferage
3. Cost of material handling
4. Number of handlingS

Techniques

1. Operational research
2. Material flow analysis
3. Computerized material retrieval system
4. ASRS (Advanced Storage & Retrieval System)

5. Packaging

Flow of Actions

1. Packaging requirement for the material (Refrigeration, Fragile etc.)
2. Primary packaging
3. Secondary packaging
4. Cost of packaging
5. Transportation requirement for packaging (Vibration proof, water or moisture tight)

Important Factors

1. Protection to product
2. Holding the product
3. Communicating the message to customers
4. Customer requirement for packaging
5. Reverse logistics for packaging
6. Recycling of packaging material
7. Cost of packaging

Techniques

1. Standardized box packaging
2. Containerization of packaging
3. Direct part marking
4. ISO 14001
5. Recycling of packaging materials
6. Reusable packaging materials
7. Eco-friendly packaging materials
8. Bar coding
9. Bumpy bar coding

10. GPS tracking system

11. RFID

6. Transportation

Flow of Actions

1. Mode of transportation

2. Cost of product

3. Speed of transportation

4. Ambiance requirement of material (Refrigeration, Vacuum)

5. Cost of transportation

6. Urgency of the product to customers

Important Factors

1. Urgency of the product

2. Cost of product

3. Cost of transportation

Techniques

1. Containerized transportation

2. Cool Chain Transport (Refrigerated Vans/Containers)

3. Multi-modal Logistics

4. Milk Run Distribution systems

5. Cross Docking

6. Direct Shipment

7. Warehousing

Flow of Actions

1. Location of the warehouse
2. Inventory level at the warehouse
3. Storage requirement of the product
4. Packaging and repackaging requirement of the product
5. Shelf life of the product

Important Factors

1. Availability of space
2. Availability of proper material handling systems
3. Strategic location
4. Packing and Re-packing facilities
5. Information and allied services

Techniques

1. Third Party Logistics
2. Third party warehousing

8. Customer Service

Flow of Actions

1. Contractual services offered to client
2. Type of customer service required for the product
3. Location of the service centre
4. Service level at the service centre
5. Cost of service vs. replacement

Important Factors

1. Contractual requirement of customer service
2. Service quality

3. Reverse logistics

Techniques

1. AMC (Annual Maintenance Contracts) and free replacements
2. Limited (free) trial period
3. Guarantee & warrantee
4. User clubs
5. Help lines, toll free number, call centers
6. CRM

The Logistical Value Proposition

Logistics Value Proposition is the assurance the entity gives to its customers concerning the expected value of the logistics function. It should be up to the customer's expectations and give a cost-benefit to the entity also. It is done in the following ways:

1. Commitment of resources to logistics department- Allocate vehicles, goods, labor, and the technology required for quick delivery and real-time tracking.
2. Cost Reduction by clubbing various orders to lessen the freight and shipping costs.
3. Setting up warehouses at various geographical locations to access the products faster without waiting for the delivery from the main factory.
4. Set up a system of issuing periodic updates of delivery to the customer to ensure customer retention.
5. Assign case managers either geographically or product-wise for quick turnaround time for resolving delivery issues.

Logistical Value Proposition Elements

Logistics should be managed as an integrated effort to achieve customer satisfaction at the lowest total cost.

The key elements that need consideration are:

1. Service Benefits:

Almost any level of logistical service can be achieved if a firm is willing to commit the required resources. In today's operating environment, the limiting factor is economics, not technology. Few examples relating to this factor are:

- Maintaining a dedicated inventory in close geographical proximity to a major customer.
- Holding a fleet of trucks in a constant state of delivery readiness.
- Maintaining dedicated communications on a real-time basis between a customer and a supplier's logistical operations.

Given this high state of logistical readiness, a product or component could be delivered within minutes of identifying a customer requirement. Availability is faster when a supplier agrees to consign inventory on-site at a customer's facility, eliminating the need to perform logistical operations when a product is needed.

The key strategic issue is how to outperform competitors in a cost-effective manner. Creation and basic logistical performance is measured in terms of:

- Availability
- Operational Performance
- Service Reliability

2. Cost Minimization:

The focus of logistics is a relatively new theory and a practice derived from a study carried out by Lewis, Culliton and Steele in 1956 in an effort to provide a new perspective concerning high-cost air transport. This implemented study was to achieve the lowest possible cost for each individual function of logistics, with little or no attention to integrated total cost trade-offs in combined costs. However, the implementation of effective logistical process costing remains a 21st century challenge.

3. Logistics Value Generation:

The key to achieving logistical leadership is to master the art of matching operating competency and commitment to key customer expectations and requirements. This customer commitment – in an exacting cost framework known as Logistics Value Propositions – is a unique commitment of a firm to an individual or selected customer groups.

What is expected is to develop and implement an overall logistical competency that satisfies customer expectations at realistic total cost expenditure. A well-designed logistical effort must have high customer impact, while controlling operational variance and minimizing inventory commitment, and, most of all, it must have relevancy to specific customers.

LOGISTICS IN ORGANIZATIONAL STRUCTURES

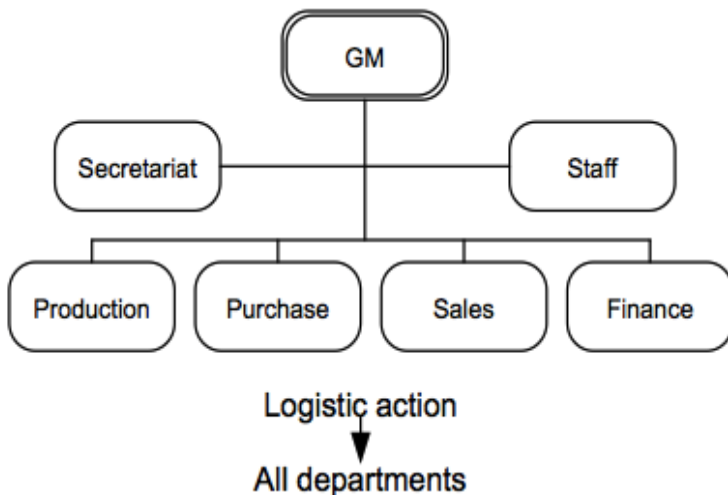
Linkages and Relationships between Purchasing and Logistics

Relationship of purchasing and logistics takes many forms. There are four basic systems linking procurement

and logistics in organizational structures mainly manufacturing companies and related modifications.

The first organizational structure is a classic functional organization structure, in which major segments are a production, purchase, sales, human resources, and finance. A logistics is not centrally managed; logistics processes are controlled by single major sections. All input logistics (ordering, shipping, warehousing of components and input material) is controlled by purchase, see fig. 1. This type of organizational structure can be found in some traditional industrial companies primarily in engineering, for example, the company Tawesco Ltd.

Figure 1: Classic organizational structure

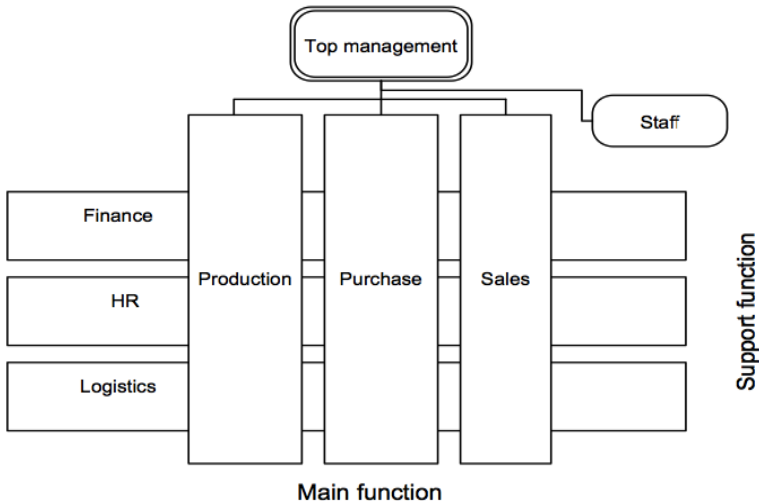


The second case, which is not very common, it is the opposite of the first. Logistics department is placed in the main section of the frontline under the direct leadership, and

a purchase is incorporated in logistics. An example of such an arrangement can be found at the company ŽDAS.

The third case linking procurement and logistics is located in a functionally oriented matrix arrangement. Purchase Department is located between the vertically actuated implementing sections of the main functions and logistics segment is classified as horizontally positioned interlocking sections to service function. Everything is shown in fig. 2. In this case purchasing and logistics overlap and cooperate together on an equal basis.

Figure 2: Functionally oriented matrix arrangement of Purchasing and Logistics

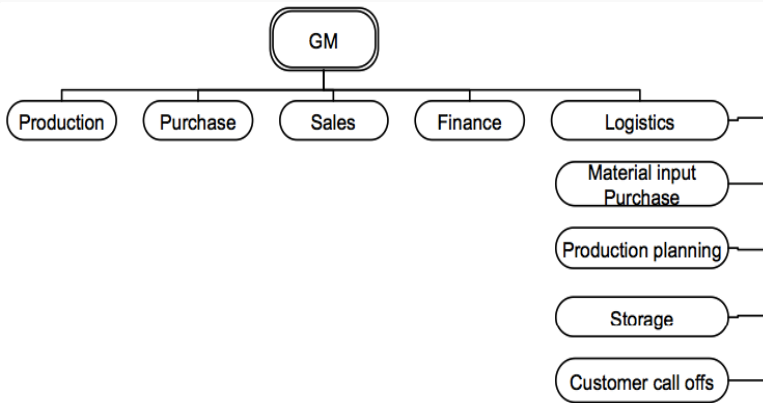


In the fourth case is a situation where on the first organizational level represented both sections. Regarding the activities section of the purchase is in charge of strategic and non-manufacturing purchasing, logistics operative then solves the purchase of materials and components for production. This model is widely used by mid-sized manufacturing companies in the automotive industry. Such

an arrangement of logistics and purchasing is for example in a company AGC Automotive Czech Inc.

Logistics of Manufacturing Company in the Organizational Structure

What role of Logistics in organizational structures of companies compared to other sectors is very variable? Almost each company has a different concept of logistics. Exceptions are companies operating in the automotive industry where logistics sections are similar in structure due to involvement in the united supply chain. Trends in the status of logistics in organizational structures are clear. In the past mainly manufacturing companies did not have separate sections of logistics and logistics were provided by other sections. The purchasing department provided orders, transport and components storage, manufacturing division ensured the production planning and internal logistics and sales department was in charge of the warehouse finished goods and transportation of finished goods to customers. This organizational structure, when the logistics are not centrally managed, belongs among functional organizational structures and some companies still, use it. At present increasing companies where logistics is centralized (Fig. 3), the logistics division reports directly to CEO.

Figure 3: Centrally managed logistics

It is because logistics costs about total costs rise, companies are involved in the supply chain, and ever-evolving information technologies unite logistical flows. It can say that along with marketing logistics are among the areas that are increasingly perceived in the organizational structures and management companies. Centralized Logistics manages the flow of materials throughout the whole company. Logistics workers take customer orders, enter requests into the production plan, order components based on production plan, they manage the input material stores and final production warehouses. logistics organizational structures

Competences in Centralized Logistics

As regards, the distribution of competences at the output of the company, logistics is in touch with the customer regarding operational requirements. Then sales department resolves a long-term customer requirements and agreements with them. Logistics cooperates with the purchase at the input in the company. Logistics sends contractors operational requirements arising as input material needed for the production plan, a purchase searches

for suppliers, conclude contracts and bargains with suppliers in case of disputes and delivery problems. One of the topics is the in-house logistics, unambiguous definition of competencies between logistics and production handling, but it is not the subject of this paper. logistics o

Types of Information Flow in Logistics Sector

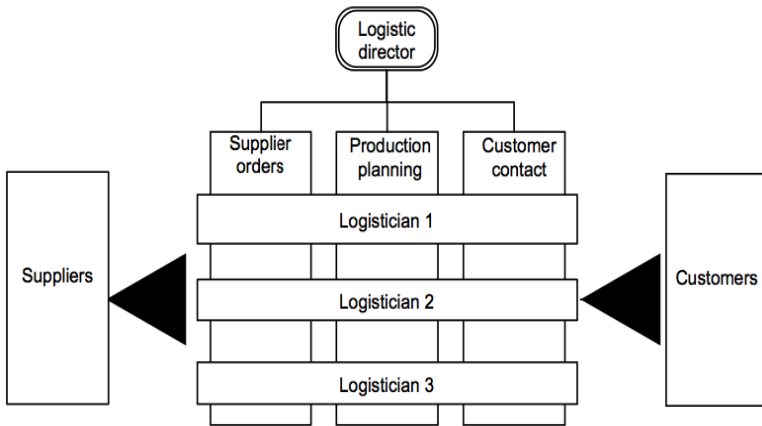
Two types of information flow in logistics sector exist:

- Information flows through projects,
- Information flows through departments.

1. Logistics Divided by Projects

Information flows and responsibilities at the centrally managed production logistics can operate in two ways. Especially small and medium-sized businesses that do not have a portfolio of hundreds of finished products and also purchased items, and the flow of production is not very difficult; they have logistics divided by projects. This means that the logistic section has several logistic specialists. Each logistic specialist cares for several customers. The logistic specialist takes orders from customers, requests; he enters requests into the production plan and orders components for their projects from suppliers (Fig. 4). The advantage of this competence distribution is operational approach and lean logistics. In the event of an economic crisis and decline in orders from the customer is responsible logistic specialist able to respond quickly and immediately to decrease supplier orders.

Figure 4: Logistics dividend by projects

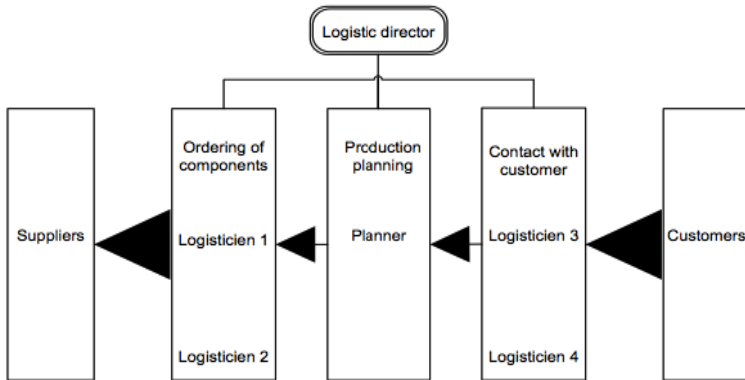


The disadvantage may be that some contractors supplying components for multiple projects and they can receive orders from one customer from multiple sources from different logistic specialists, which can be confusing. logistics organizational structures

2 Information Flow Through Departments

Larger companies with a diverse portfolio of finished goods or input components prefer logistics flows through departments. Business logistics department receives requests from all customers, forwards them to the production planning department that creates a plan which transmits information to input logistics department, and it orders and controls inputs of materials and components (Fig. 5). In this case, it is extremely important the proper flow of information, data and control mechanisms to prevent any inaccuracies and loss of information in the logistics flow.

Figure 5: Information flow through departments



SUPPLY CHAIN SYNCHRONIZATION

Supply chain synchronization is an ecosystem of connected, collaborative data partners, whereby information gets collected, analyzed, and utilized in real time. All stakeholders within the critical path obtain accurate visibility, identify weaknesses, streamline processes, and mitigate risk.

Supply chain synchronization requires for the collection, analysis, and use of information to take place in real time to enable the management of the critical path order life cycle with pinpoint accuracy to mitigate risk so that:

- suppliers can organize raw materials and plan production around realistic and quantified lead times
- logistics providers can determine shipping requirements with certainty

- distribution centers have visibility for receiving and planning, labor and stock holding optimization
- shop floor optimizers get to understand when product is due to store
- finance departments know with certainty when to trigger payments and hedge currencies, etc.

In essence, all participants in the supply chain must work together, knowing their contribution interconnects with others.

The Five Steps to Achieve Supply Chain Synchronization

The five steps to achieve supply chain synchronization include the following:

- **Collaboration and Engagement Facilitation** - Collaboration and engagement facilitation are key within the first step of supply chain synchronization. You must enlist supply chain participants as stakeholders and ensure that everyone is engaged within their role in making production run as smoothly as possible, both internally and externally. You must segment your suppliers and identify any relationships that yield profits and begin the process with them. Without everyone on the same page, your supply chain will be unsynchronized and overall inefficient, which is why it is important to ensure that everyone understands their part within the process.
- **Visibility Establishment** - Thorough visibility allows you to make decisions based off of all of the information provided. You do not want cloudy information that is hard to understand or is missing important components/factors,

which is why visibility establishment is an absolute must for supply chain synchronization. Map your end-to-end processes among various departments and ensure that you identify and road test every process to provide exact lead times with all considerations and known buffers. Share the data with everyone and you will easily enhance visibility within your supply chain.

- **Collecting Real-Time Data** - It is important to define who provides data, how you can capture it, and where it needs to go in order for all parties to work off one version of the truth. Collecting real-time data is an integral step in supply chain synchronization.
- **Monitoring, Managing, and Executing** - Monitoring, managing, and executing ensures that the system informs everyone in real time on any deviation to the plan or where disruption may loom. This enables all participants to collaborate and take action where it is needed.
- **Measuring, Remodeling, and Re-Engineering** - It is important to measure your supply chain synchronization efforts regularly instead of waiting until the end of the year. This allows you to continuously examine whether or not your efforts are effective and if you should continue in the direction that you are heading. Therefore, let the system analyze and predict to help plan for future ordering for your production process.

Transportation functionality

Transportation is one of the most visible elements of logistics operations. As consumers, we are accustomed to seeing trucks and trains moving products or parked at a distribution facility. While this experience provides a good visual understanding of transportation elements, it does not allow the necessary depth of knowledge to understand

transportation's role in logistics operations. This section establishes that foundation by reviewing functionality provided by transportation and the underlying principles of transport operation.

Following are the two major functionality of Transportation

- Product movement
- Product storage

Product Movement

Whether the product is in the form of materials, components, assemblies, work-in-process, or finished goods, transportation is necessary to move it to the next stage of the manufacturing process or physically closer to the ultimate customer. A primary transportation function is product movement up and down the value chain. Transportation utilizes temporal, financial, and environmental resources, it is important that items be moved only when it truly enhances product value.

The major objective of transportation is to move product from an origin location to a prescribed destination while minimizing temporal, financial, and environmental resource costs. Loss and damage expenses must also be minimized. At the same time, the movement must take place in a manner that meets customer demands regarding delivery performance and shipment information availability.

Product Storage

A less common transportation function is temporary storage. Vehicles make rather expensive storage facilities. However, if the in-transit product requires storage but will be moved again shortly (e.g., in a few days), the cost of unloading and reloading the product in a warehouse may

exceed the profitability. A second method to achieve temporary product storage is diversion. This occurs when an original shipment destination is changed while the delivery is in transit. Traditionally, the telephone was used to direct diversion strategies. Today, satellite communication between enterprise headquarters and vehicles more efficiently handles the information.

Principle of Transportations

Economy of scale

It refers to the characteristic that transportation cost per unit of weight decreases when the size of the shipment increases. For example, truckload (TL) shipments (i.e., shipments that utilize the entire vehicle's capacity) cost less per pound than less-than-truckload (L TL) shipments (i.e., shipments that utilize a portion of vehicle capacity). It is also generally true that larger capacity transportation vehicles such as rail or water are less expensive per unit of weight than smaller capacity vehicles such as motor or air. Transportation economies of scale exist because fixed expenses associated with moving a load can be spread over the load's weight. As such, a heavier load allows costs to be "spread out," thereby decreasing costs per unit of weight. The fixed expenses include administrative costs of taking the transportation order, time to position the vehicle for loading or unloading, invoicing, and equipment cost. These costs are considered fixed because they do not vary with shipment volume.

Economy of distance

It refers to the characteristic that transportation cost per unit of distance decreases as distance increases. For example, a shipment of 800 miles will cost less than two shipments (of the same combined weight) of 400 miles.

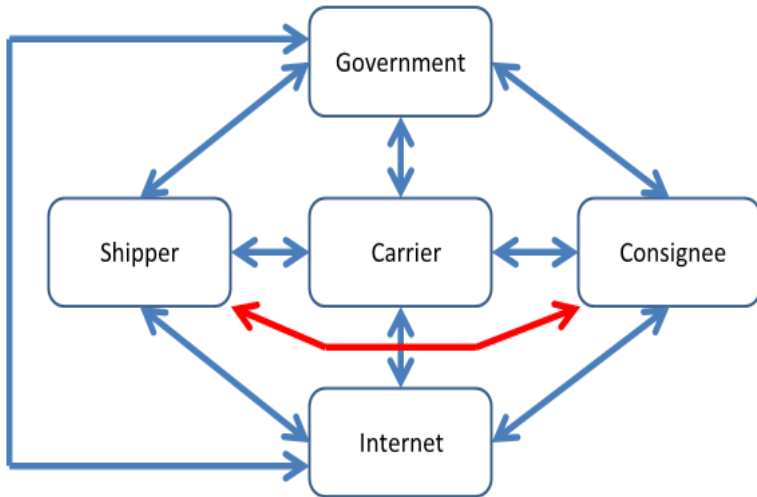
Transportation economy of distance is also referred to as the tapering principle since rates or charges taper with distance. The rationale for distance economies is similar to that for economies of scale. Specifically, the relatively fixed expense incurred to load and unload the vehicle must be spread over the variable expense per unit of distance. Longer distances allow the fixed expense to be spread over more miles, resulting in lower overall per mile charges.

Transport Participants

The transportation environment impacts the range of decisions that can be implemented in a logistical system. Unlike most commercial transactions, transportation decisions are influenced by six parties:

- Shipper
- Destination party traditionally called the consignee.
- Carriers and agents
- Government
- Internet
- The public

Following Figure illustrates the relationship among the involved parties. To understand the complexity of the transportation environment it is useful to review the role and perspective of each party



Shipper and Consignee

The shipper and consignee have a common interest in moving goods from origin to destination within a given time at the lowest cost. Services related to transportation include specified pickup and delivery times, predictable transit time, and zero loss and damage as well as accurate and timely exchange of information and invoicing.

Carrier Agents

The carrier, a business that performs a transportation service, desires to maximize its revenue for movement while reducing associated costs.

As a service business, carriers want to charge their customers the highest rate possible while minimizing labor, fuel, and vehicle costs required to complete the movement.

To achieve this objective, the carrier seeks to coordinate pickup and delivery times in an effort to group or consolidate many different shippers' freight into movements that achieve economy of scale and distance.

Brokers and freight forwarders are transport agents that facilitate carrier and customer matching.

Government

The government has a vested interest in transportation because of the critical importance of reliable service to economic and social well-being. Government desires a stable and efficient transportation environment to support economic growth.

A stable and efficient transportation environment requires that carriers provide essential services at reasonable cost. Because of the direct impact of transportation on economic success, governments have traditionally been more involved in the practices of carriers than in most other commercial enterprises.

Internet

A recent development in the transportation industry is a wide assortment of Internet-based especially cloud computing services. The primary advantage of such communications is the ability of carriers to share real time information with customers and suppliers.

Information exchange operating over the Internet provides carriers the opportunity to aggregate their purchasing and identify opportunities across a wide range of potential vendors.

The availability of real time information is improving shipment visibility to the point where tracing and tracking are no longer a challenge. In addition to real time visibility, the Internet can be used to share information concerning scheduling and capacity planning.

Transport Cost Characteristics

Transportation usually represents the most important single element in logistics costs for most firms. Freight movements have been observed to absorb between one-third and two-thirds of total logistics costs.

Variable and Fixed Costs

A transportation service incurs a number of costs, such as labor, fuel, maintenance, terminal, roadway, administrative, and others. This cost mix can be arbitrarily divided into those costs that vary with services or volume (variable costs) and those that do not (fixed costs).

Fixed costs are those for roadway acquisition and maintenance, terminal facilities and transport equipment.

Variable costs usually include line-haul costs such as fuel and labor, equipment maintenance, handling, and pickup and delivery.

Cost Characteristics by Mode

The type of services that a carrier is likely to emphasize is indicated by the nature of the general cost function and by the relationship of the mode to those of other carriers.

1 Rail

As a transporter of freight and passengers, the railroad has the characteristics of high fixed cost and relatively low variable cost.

Loading, unloading, billing and collecting, and yard switching of multiple-product, multiple-shipment trains contribute to high terminal costs for rail.

Increased per-shipment volume and its effect on reducing terminal costs result in some substantial

economies of scale, that is, lower per-unit costs for increased per-shipment volume.

2 Water

The major capital investment that a water carrier makes is in transport equipment and, to some extent, terminal facilities.

Waterways and harbors are publicly owned and operated. Little of this cost, especially for inland waterway operations, is charged back to water carriers. The predominant fixed costs in a water carrier's budget are associated with terminal operations.

Terminal costs include the harbor fees, as the carrier enters a seaport, and the costs for loading and unloading cargo.

Water is one of the least-expensive carriers of bulk commodities over long distances and in substantial volume.

3 Air

Air transportation has many of the same cost characteristics as water carriers. Air terminals and the air space are generally not owned by the airline companies.

Airlines purchase airport services as needed in the form of fuel, storage, space rental, and landing fees.

If we include ground handling and pickup and delivery in the case of airfreight operations, these costs are the terminal costs for air transportation.

Combined fixed and variable expenses generally make air transportation a premium service, especially for short distances; however, distribution of terminal expenses and other fixed charges over increased volume offers some reduction in per-unit costs.

Substantial per-unit cost reductions come from operating aircraft over long distances.

Carrier Pricing Strategies

When setting rates to charge shippers, carriers typically follow one or a combination of two strategies.

Although it is possible to employ a single strategy, the combination approach considers trade-offs between cost of service incurred by the carrier and value of service to the shipper.

1 Cost-of-Service

The cost-of-service strategy is a build-up approach where the carrier establishes a rate based on the cost of providing the service plus a profit margin.

For example, if the cost of providing a transportation service is \$200 and the profit markup is 10 percent, the carrier would charge the shipper \$220.

2 Value-of-Service

It is an alternative strategy that charges a price based on value as perceived by the shipper rather than the carrier's cost of actually providing the service. For example, a shipper perceives transporting 1000 kg of electronics equipment as more critical or valuable than 1000 kg of coal since electronics are worth substantially more than the coal.

As such, a shipper is probably willing to pay more for transportation. Carriers tend to use value-of-service pricing for high-value goods or when limited competition exists.

Value-of-service pricing is illustrated in the premium overnight freight market.

When FedEx first introduced overnight delivery, there were few competitors that could provide comparable service, so

it was perceived by shippers as a high-value alternative. They were willing to pay \$22.50 for overnight delivery of a single package. Once competitors such as UPS and the United States Postal Service entered the market, rates dropped to current discounted levels of \$5 to \$10 per package. This rate decrease more accurately reflects the value and cost of this service.

Total Cost Concept & Trade-offs

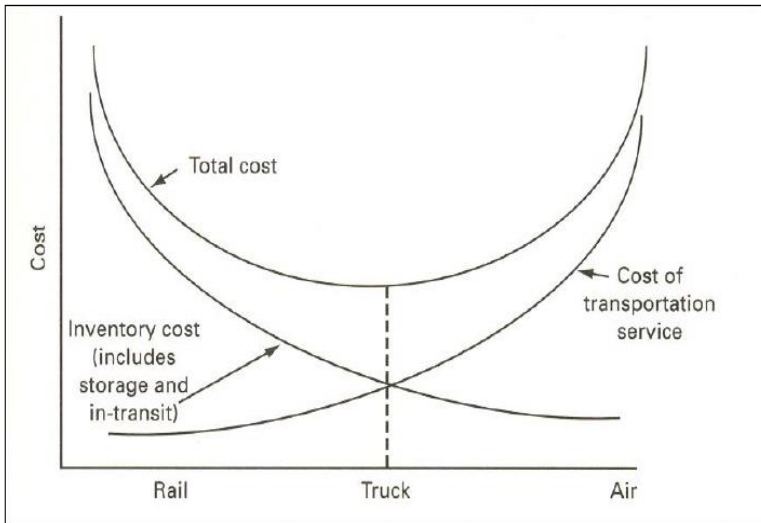
Central to the scope and design of the supply chain and logistics system is trade-off analysis, which, in turn, leads to the total cost concept.

Choosing a transportation service on the basis of either lowest rates or fastest service may not be the appropriate method. Therefore, the basic problem in supply chain is one of cost-conflict management.

Wherever there are substantial cost conflicts among activities, they should be managed in a coordinated manner.

Following Figure shows that when a transportation service is being selected, the direct cost of the transport service and the indirect cost effect on inventory levels in the logistics channel due to different delivery performance of carriers are said to be in cost conflict with each other.

The best economic choice occurs at the point where the sum of both costs is lowest.



Cost Trade-off between Transportation and Inventory Costs as a Function of Transportation Service Characteristics

The total cost concept applies to more than the problem of selecting transportation service.

However, at times, decisions made by a firm in a channel of distribution affect the logistics costs of another firm. For example, the inventory policies of a buyer affect both the inventory costs of the shipper and the operating costs of the carrier.

TRANSPORT ADMINISTRATION

Transportation administration/ management is all about finding the most optimal source of transportation for product carrying. This process involves managing all inbound and outbound transports. Transportation management incorporates the overall network design of the carrier and its level of flow. With an effective transportation management

strategy in motion, your clients will take in the benefits of more stability, visibility, and control of their transportation costs and functions.

Responsibilities of transport administrator

- Coordinating with the Transportation team regarding transportation planning and strategies
- Reviewing customer requirements and suggesting vehicle types and freight charges
- Ensuring compliance with the safety standards and transportation department SOPs
- Handling any customer complaints and processing the urgent ones to the Senior Management
- Ensuring the safe and timely delivery of goods
- Coordinating loading and unloading activities
- Scheduling vehicle maintenance appointments
- Receiving customer requests and bookings
- Planning daily transportation schedules and assigning drivers
- Performing administrative duties such as preparing and maintaining shipment documents, invoices, and logbook entries
- Liaising with drivers for smooth movement of goods and materials

- Developing and implementing new and improved transportation procedures
- Updating freight charges periodically
- Managing transportation budget and checking for newer/shorter routes
- Assisting in training new recruits
- Monitoring refueling of vehicles

TRANSPORT DOCUMENTS

A transport document is a kind of document used to convey information about cargo that is being transported. They are contracts for carriage of goods exchanged between different actors. They differ depending on the method of carriage used.

Bill of Lading

The Bill of Lading (B/L) is a contract for maritime carriage that specifies the taking of responsibility, or the loading, of goods by the carrier. It contains detailed information on the goods, the boat and the port of destination.

It is a document of title to the goods, that gives right to their ownership. The carrier commits to delivering the goods to the person who holds the bill of lading and who will come and get the goods. It is a transferable title: the originals may be transferred by endorsement.

Sea Waybill

The Sea Waybill documents only the carriage contract signed by the shipper and the maritime transport line and represents the receipt of the goods. It contains

detailed information on the port of embarkation, the port of destination, the name of the ship, the name of the shipper, the name and the address of the recipient.

Different from the Maritime Bill of Lading, the Sea Waybill is not a document of title to the goods and is non-transferable.

Consignment Note (CMR)

The CMR consignment note certifies the **carrier's taking of responsibility** for the merchandise and shipping upon signature by the carrier. It documents a carriage contract of the goods that exists independently of them. It must be established in the form specified by the "Convention on the Contract for the International Carriage of Goods by Road" (CMR).

Air Waybill (AWB)

The Air Waybill constitutes the proof of the carriage contract, the taking of responsibility for the goods and the documentation of the prices. It certifies the effective shipment once the carrier has entered the date and number of the flight. It is standardized by the IATA (International Air Transport Association), that moreover, develops solutions for the demineralization of AWB's.

In the case of grouping, when goods are gathered with other goods to complete a load for the same destination, the forwarder that groups the goods becomes legally, with respect to the carrier, the shipper and thus the drafter of the parent air waybill, referred to as the Master Air Waybill (MAWB). A "daughter" air waybill referred to as the House Air Waybill (HAWB) documents the contract between the forwarder and each of the shippers included in a grouping.

Rail Consignment Note (CIM)

The Rail Consignment Note (CIM) or International Consignment Note is a document governed by the 1980 Convention concerning International Carriage by Rail (COTIF-CIM). It is issued by the carrier and is considered as the **rail carriage contract**. This document is non-transferable and is not requested upon arrival.

Module V

INTERNATIONAL LOGISTICS AND SUPPLY CHAIN MANAGEMENT

Logistics refers to the services or activities required for the management of physical movement of goods from the source of origin to the source of destination. It involves transaction channel, distribution channel and communication channel for faster and efficient provision of logistics services. International Logistics, therefore, refers to the integration and management of activities including inventory databases and shipping schedules, material handling, production, packaging, inventory, transportation, distribution, storage, and security for the resources of organizational supply chains beyond the political boundaries of the nations involved in the global trade.

With the advent of liberalization and glottalization of world trade, international logistics is becoming more and more important to maintain the efficient, effectiveness and value addition in the international trading activities. In this highly competitive world trade market, the business entities need to take deep insights into global trade scenario and also required to think globally with an aim to manage its supply chain network. All this is possible with the adoption and application of international logistics at wider level by these companies. In addition, international logistics are extremely important to achieve the marketing and overall broader objectives of a firm.

Further, the concept of logistics, and in our case, international logistics, is gaining importance with the passage of time on account of following reasons:

- Increasing transportation cost
- Achieving product efficiency at higher level
- Changing perspectives of inventory management
- Proliferating product lines
- Increasing uses of Information and Communication Technology
- Increasing variations in customer needs and demands
- Relaxing rules and regulations
- Strengthening distributional networks

Objectives of International Logistics

The basic objective which a company tries to achieve through using the logistics at global level or even at national level is to maintain the ensured flow of goods & services from seller to the buyer with the following:

- Right product
- Right quantities and assortments
- Right places
- Right time
- Right cost/price
- Right condition

Moreover, logistics management starts with as considering the customer needs till its fulfillment through product supplies in line with all the aspects of performance

including arrangement of the inputs, manufacture of the goods and the physical distribution of the products. However, there are some definite objectives to be achieved through a proper logistics system which are as follows:

1. Improving Customer Service

The effective management of international logistics system can help in improving the level of customer service by developing an effective system of warehousing, quick and economic transportation, along with maintenance of optimum level of inventory because the level of service directly affects the cost of physical distribution.

2. Increased Customer Response

Now-a-days, the information & communication technology has increased the capability to postpone logistical operations to the latest possible time and then achieve the faster delivery of required inventory, thereby, resulting in the elimination of excessive inventories traditionally stocked in anticipation of customer requirements.

3. Reduction in Distribution Costs

The distribution cost is comprised of various elements, like, transportation, warehousing and inventory maintenance, and any reduction in the cost of one the elements are likely to increase the cost of the other elements. Thus, the objective of the firm should be the reduction of the total cost of distribution and not just the cost incurred on any one of the elements. To serve this purpose, the total cost of alternative distribution systems should be analyzed and the one with the minimum total distribution cost should be selected.

4. Increasing Sales

The physical distribution/logistics system in a firm also aims to generate more sales. A firm can attract more customers by offering better services at lower prices. For instance, by introducing decentralization in its warehousing operations or by using economic and efficient modes of transportation, a firm can achieve larger market share. Further, by avoiding the out-of-stock situation, the loss of loyal customers can be curtailed.

5. Creating Time & Place Utilities

The logistical system also focuses on creating time and place utilities to the products & services. Till the products are physically moved from the place of their origin to the place of destination, they do not serve any purpose to the users. In the same manner, the products & services have to be made available at the time when they are needed for consumption. Moreover, a rapid mode of transportation should be selected to move the products from one place to another in the shortest possible time. Therefore, time and place utilities can be created in the products & services by adopting an efficient system of physical distribution.

6. Stabilizing Product Prices

The international logistics system also aims at stabilizing the prices of the products. It can be achieved by controlling & maintaining the flow of the products to the market through a judicious use of available transport facilities and compatible warehouse operations. To say, maintaining the stock of the raw material during the periods of excess supply and making it available during the periods of short supply, the prices can be stabilized.

7. Quality Management

The international logistics system is a significant part & parcel of developing and maintaining continuous improvement in total quality management. The long-term objective of the international logistics system is to seek continuous quality improvement. Overall commitment to total quality management is one of the major forces responsible for the logistical renaissance because the logistical costs cannot be reversed, if they are increased once. In fact, when the quality fails to match the set standards, the logistical performance typically needs to be reversed and then repeated. Moreover, logistics itself must perform to demanding quality standards.

8. Supporting Product Life-Cycle

A good logistical system helps to support the life cycle. Product recall is a vital competency resulting from increasingly rigid quality standards, product expiration dating and liability for unfavourable consequences as well. The return logistics requirements are also a product of the increasing number of regulatory norms & laws prohibiting disposal and encouraging recycling of beverage containers as well as packaging materials. Further, the operational requirements of reverse logistics have stretched their scope from lowest total cost, to say, returning bottles for recycling, to maximum performance solutions for critical recalls. The critical point is that the sound logistical strategy can't be formed without careful review of reverse logistical requirements.

9. Consolidating Transportation Costs

Transportation cost is directly linked to the type of product, size of shipment, and distance which the concerned shipment is supposed to travel. Many Logistical

systems which are equipped with the premium services depend on high-speed, small shipment transportation because the larger the overall shipment and the longer the distance it is to be transported, the lower the transportation cost per unit. This needs innovative initiatives aimed at clustering the small shipments for consolidated movement which are required to be facilitated by the working arrangements that transcend the entire supply chain.

Scope of International Logistics

With the continuously expanding scope of international logistics, a number of activities and services are now included in the periphery of the concept. However, following areas are majority considered as a part & parcel of international logistics:

- Forecasting Customer Demands
- Strengthening Communication Channels
- Maintaining Distribution Channels
- Controlling Inventory Requirements
- Handling Material Requirements
- Processing Orders
- Selecting Location for Plant & Warehouse
- Procuring Raw Materials
- Managing Packing & Packaging Requirements
- Regulating Traffic & Transportation
- Creating Time & Place Utilities
- Movement Consolidation
- Handling Reverse Logistics

- Maintenance of Customer Service & Support

Elements of International Logistics

1. Processing Customer Orders

The first element of the international logistics system is the processing of orders of the customers. With an aim to provide faster customer service, the orders received from customers need to be processed within the least possible time. This element comprises of the activities like, receiving the order, recording the order, filling the order, and assembling all such orders for transportation, etc. The company and the customers will be benefited when these steps are carried out quickly and accurately. The error committed at this stage, generally, can prove to be very costly. To say, if a wrong product or the same product with different specifications is supplied to the customer, it may lead to cancellation of the original order which will increase the cost of reverse logistics and can also harm the company's goodwill at the same time.

Similarly, if the order is not executed within a reasonable time, it may lead to serious consequences. Further, the high speed data processing techniques are presently available leading for the rapid processing of the orders.

2. Storage and Warehousing

This is the second element which involves assorting products in order to create time utility. The major objective of the storage & warehousing activities is to make proper arrangements for the placement of goods, providing storage facility to store them, consolidating them with other similar products, splitting up them into smaller quantities and building up assortment of products. Generally, more the number of warehouses a firm has, the lesser would be the

time taken in serving customers at different locations. However, this would result in increased storage & warehousing cost. Therefore, the firm has to maintain a balance between the cost of warehousing and the level of customer service.

3. Managing & Controlling Inventories

The next element involves decision-making process regarding managing and controlling inventories which is important for the success of physical distribution. This element becomes more important in case the cost of inventories is higher. Moreover, the new concept of Just-in-Time-Inventory decision is increasingly becoming popular with a number of companies, now-days.

The decision regarding level of inventory involves estimated demand for the product. A correct estimate of the demand helps to maintain requisite inventory level and control the inventory costs as well. It helps the firm in terms of the cost of inventory and supply to customers in time and also to maintain production at a consistent level. The major forces determining the inventory levels include the company's policy regarding the customer service level, degree of accuracy of the sales forecasts, responsiveness of the distribution system, so on so far. The cost of inventory consists of holding cost (such as cost of warehousing, tied up capital and obsolescence) and replenishment cost (including the manufacturing cost).

4. Transportation

The element of transportation consists of moving goods from points of production and sale to points of consumption in the quantities required at times needed and at a reasonable cost. The transportation system increases time and place utilities to the goods handled and

therefore, increases their economic value. With an aim to achieve these goals, transportation facilities must be adequate, regular, dependable and equitable in terms of costs and benefits of the facilities and service provided.

5. Information Monitoring

The information monitoring element of international logistics system requires the logistics' handling personnel continuously updated information about inventory, transportation and warehousing. For instance, with respect to inventory, the information about the present stock position at each location, future commitment and replenishment capabilities are required on an ongoing basis. In the same manner, before choosing a carrier, the information about the availability of relevant modes of transportation, their costs, services and suitability for a particular product is needed. In case of warehousing activities, the information with respect to space utilization, work schedules, unit load performance, etc., is required. For receiving all this information, an efficient management information system would be of immense use in controlling costs, improving services and determining the overall effectiveness of distribution. Further, it is difficult to correctly assess the cost of physical distribution operations. However, if the correct information is available, it can be analyzed & processed systematically and a great deal of saving can be ensured.

6. Support Facilities

The support facilities element of international logistics system is comprised of a number of planning activities which are, usually, directed toward ensuring that all required permanent or transitory operating and support facilities, including training, field and depot maintenance, storage, operational, and testing are available

concurrently with system fielding. Planning must be comprehensive and include the need for expansion, diversification and modification of the existing facilities as well as creating new facilities. It also takes research & developmental activities under its purview with an aim to define and establish impacts on life cycle cost, funding requirements, facility locations and improvements, space requirements, environmental impacts, duration or frequency of use, safety & security requirements and health standards, and security restrictions. Furthermore, it also includes any utility requirements, for both fixed and mobile facilities, with thrust on containing requirements of scarcely available resources.

Importance of logistics in global economy

The success of any business or economy of the country in global markets is also depending on the importance of trade logistics solutions. Let's discuss 7 contemporary trends that can be identified regarding the importance of logistics in global industry.

- **Generation of Demand**

The demand of any product is improved significantly from increased mobility, unobstructed movement of products & services and access to better logistics infrastructure. This is because enhanced trade and logistics infrastructure create place, time and form utilities for the customers & users. Both customers and users can be serviced at any time and at any place. Thus, improved international logistics infrastructure helps in increasing the overall sales of the company's products.

- **Cost Reduction in Doing Business**

Improved logistical infrastructure helps in keeping cost of business at the lower side as transportation of

products from one place to another becomes almost uninterrupted due to better ports, railway network, roads and civil aviation infrastructure.

For example, Due to better road connectivity in China, a truck can travel 1,300 km into this country in about 74 hours. And the same distance, which is equivalent to distance between Delhi-Kolkata, is covered in about 144 hours in India. This delay not only extends trade cycle, but the quality of certain goods get poor and fetches lower prices in markets.

- **Tapping Clients in the World**

Improved global logistics services from top logistics companies and better transportation are important for the movement of goods and services from one region to another. This helps companies to have a tap on the customers in every nook and corner of the world.

For example, Indian industry has many potential fields such as electronics, engineering, chip designing, auto components, etc. It can contribute to the world's markets only if the country has improved trade logistics infrastructure and networking systems; otherwise the business opportunities can be outpaced by the nation's rivals from other developed countries.

Hence, any country needs to have quality logistics infrastructure to tap clients all over the world.

- **Ensuring Rapid Economic Growth**

In the development process of any country, growth in the economy plays a vital role. This is possible from the expansion in trade & logistics infrastructure that create demand in economic system for products such as iron & steel, cement and manpower.

For example, India has to make its logistical infrastructure better, which will not only grow its economy but also help its companies to accomplish a sustained superior performance in international markets through enhanced trade supply chain process.

- **Bridging Gap between Demand and Supply**

How to bridge gaps between demand and supply of a product? This is one of the major challenges that any company faces in international markets at all levels from sourcing of raw materials to work in progress to distribution to customers. So, better transporting goods from one place to another and timely supply of products to meet the demand will fill the gap between demand and supply of a product.

For example, China with main economic clusters on the east coast results to transporting commodities at far-away regions in the western and remote northern parts of the country. This creates the problem of demand and supply in the country's economic system. Better connectivity from road, rail network, airstrips and sea helps companies to distribute their resources between places where there are abundant resources and where there are scarce.

- **Strategic Infrastructure for Global Integration**

Trade logistic infrastructure and transportation play an important role in conditions that affect regional, national and international economic entities of companies in accessing global markets.

For example, Nepali carpet exporters transport their goods towards the Nepal border by trucks that are unloaded for customs clearance at Birgunj in India. The products are again loaded on Indian trucks to move towards Kolkata by

road transport. The shipment is then unloaded again for loading on ship and transhipped to Singapore.

- **Ensuring Critical Supplies on Time**

An efficient logistics system in international trade operations helps companies in making timely supply of products to their international buyers. Due to complex functionality of logistic system and long distance involved between two countries, the problem of safety, care and timing of shipment often cause nightmares to suppliers, particularly in case of perishable & high value products and goods with expiry date restrictions. Such products include newspapers, flowers and marine products.

GLOBAL SUPPLY CHAIN

Global supply chains are networks that can span across multiple continents and countries for the purpose of sourcing and supplying goods and services. Global supply chains involve the flow of information, processes and resources across the globe.

Difference Between Global Supply Chain vs Local Supply Chain

A global supply chain utilises low-cost country sourcing and refers to the procurement of products and services from countries with lower labour rates and reduced production costs than that of the home country.

A global supply chain will usually flow from your own organisation in your home country as a buyer across your supplier tiers; it is these suppliers who will be located in other areas of the globe.

A local supply chain will look to optimise suppliers who are regional to your own organisation, in some instances organisations will look to leverage “home grown”

supply routes, so all suppliers feeding into your supply chain will be located within the country in which your organisation is based, or the supply chain can be even closer in to your organisation and may even be within the same state/city/district, which often gives a clearer visibility of the whole supply chain from raw material through to consumer.

However there are both positives and negatives with global supply chains and the total landed cost or total cost of ownership should always be factored into the true costs.

Supply chain integration

Supply chain integration essentially means that the information and communication systems of all stakeholders are able to seamlessly exchange information through all planning, execution and completion of transport and logistics operations throughout a product's life time.

Supply chain integration is recognized as an important factor to attain superior supply chain performance, as it offers a host of competitive advantages – including complete transparency all the way from supplier to customer.

Supply chain security

Supply chain security is the part of supply chain management that focuses on the risk management of external suppliers, vendors, logistics and transportation. Its goal is to identify, analyze and mitigate the risks inherent in working with other organizations as part of a supply chain. Supply chain security involves both physical security relating to products and cyber security for software and services.

Because supply chains can vary greatly from group to group, and many different organizations may be involved, there is no single set of established supply chain security guidelines or best practices. A complete supply chain security strategy requires following risk management principles and cyber defense in depth.

Physical supply chain security and integrity

In the past, supply chain security primarily focused on physical security and integrity. Physical threats encompass risks with internal and external sources, such as theft, sabotage and terrorism.

Organizations often mitigate physical attacks by tracking shipments and checking regulatory paperwork. Additionally, vendors could be required to secure shipments following specific quality guidelines, and a business could employ several vendors to ensure a steady supply of commodity products.

External auditors or inspectors could go to a factory, and businesses could run background checks on personnel as well. Shipments could also be logged, guarded and checked prior to and after shipping to stop tampering or theft.

Cyber threats and supply chain security

More recently, cyber threats have risen to the forefront of supply chain security concerns. Cyber threats refer to vulnerabilities in IT and software systems, like malware attacks, piracy, unauthorized ERP access and unintentional or maliciously injected backdoors in the purchased, open source or proprietary software used by organizations.

Supply chain security here primarily involves minimizing risks from using software developed by another organization, and securing organizational data accessed by another organization in your supply chain. Organizations cannot take for granted that the software that they use or purchase is secure.

Supply chain security best practices

Supply chain security touches on many areas, and will vary greatly from organization to organization. Risk management principles can guide the strategy to identify threats or potential issues and implement appropriate mitigations. No single set of best practices can cover every situation.

An in-depth defense strategy can greatly improve overall supply chain security. Best practices for supply chain security include:

- Log and track shipments. Use automated notifications for the sender and receiver.
- Use locks and tamper-evident seals during shipping.
- Inspect factories and warehouses.
- Require background checks on employees.
- Use accredited or certified suppliers.
- Perform security strategy assessments with local laws and governance policies in mind.
- Perform penetration and vulnerability testing on partners with which you share data.
- Authenticate all data transmission and identify requesters.
- Use permissions or role-based access to data.

- Require minimum cyber security or specific best practice baselines of vendors and resellers.
- Use licensed third-party auditors to certify potential partners.
- Train employees to be alert to changes and inconsistencies.
- Regularly audit open source and vendor source code.
- Restrict third-party programs' access and permissions.
- Use network level scanning, behavioral analysis and intrusion detection to identify potential breaches.
- Have a response plan in place for quickly acting on discovered threats.
- Consult governmental guidelines and regulations appropriate for your region.

By applying the mitigations best suited to its business sector, an organization can greatly improve its supply chain security posture.

International Sourcing

International Sourcing/ Global sourcing is a procurement strategy in which a business buys goods and services from international markets across geopolitical boundaries to save money by using cheap raw materials or skilled labor from low-cost countries. Global sourcing can help businesses tap into advanced skills, resources and technology not available in their home country.

Business process outsourcing (BPO) is a classic example of global sourcing. Several companies have set up their call centers in Asian countries, such as India and the

Philippines, to benefit from the availability of low-cost, skilled labor.

Over the last several years, China has led the world in manufacturing production and value. Labor costs are significantly lower in China, yet production quality is extremely high. That's why China parts sourcing has been, and continues to be, the most popular way to have component parts manufactured for many top US-based companies. However, due to the recent tariffs imposed on products from China, India product sourcing is starting to emerge as a viable alternative to China factories.

Global sourcing involves more than just searching for products globally. It is also an effort to improve certain aspects of manufacturing, such as:

- Supplier selection and performance
- Speed to market
- Estimation of product costs
- Trade compliance
- Auditing

Levels of Global Sourcing

There are five levels of global sourcing, including:

- Level 1: Domestic purchases only
- Level 2: Global purchases made on an as-needed basis
- Level 3: Global purchases included in the purchasing mix
- Level 4: Centrally-coordinated purchasing across global locations
- Level 5: Global coordination and integration with other functional groups.